



HSC Mathematics (QLD Specialist)

Matrix Practice

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*"As for everything else, so for a mathematical theory:
beauty can be perceived but not explained."*

Arthur Cayley (Founder of matrix algebra)

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1 Introduction

Remark 1.1 (Syllabus Note)

Matrices are not part of the current NSW NESA Mathematics syllabuses. The content in this booklet is adapted from the Queensland (QCAA) Specialist Mathematics syllabus. It serves as an excellent enrichment topic for students interested in linear algebra, computer graphics, and machine learning.

1.1 Project Overview

This booklet is a growing collection of matrix problems. It introduces matrix arithmetic, determinants, inverses, and applications to solving systems of linear equations and geometric transformations.

1.2 Target Audience

This resource is written for students who want to explore linear algebra beyond the standard NSW syllabus, in a clear, classroom-friendly format.

1.3 How to Use This Booklet

Read the short summary of the core formulas first, then attempt each problem before reading the solution.

1.4 Knowledge Needed

The key background knowledge is:

- Basic arithmetic and algebraic manipulation.
- Understanding of systems of linear equations.
- Basic 2D geometry (for geometric transformations).

1.5 Core Formulas and Ideas

2 Matrix Fundamentals

This section covers the essential matrix theory that underpins the worked problems and practice exercises in the rest of this booklet. Since matrices are not part of the NSW NESA syllabus, this section serves as a self-contained primer adapted from the Queensland (QCAA) Specialist Mathematics curriculum (Chapters 8 and 9 of *Cambridge Maths Unit 3 & 4 Specialist*).

Each topic presents the key ideas, followed by one or more worked examples.

2.1 Revision of Matrix Algebra

A **matrix** is a rectangular array of numbers arranged in rows and columns. Matrices are used extensively in mathematics, science, economics, and computer science to organise data, represent systems of equations, and describe geometric transformations.

Notation and Terminology

A matrix with m rows and n columns is called an $m \times n$ matrix (read “ m by n ”). We typically denote matrices by uppercase letters ($A, B, C, \dots$) and their individual entries by lowercase letters with subscripts. The entry in row i and column j of matrix A is written a_{ij} .

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}$$

Special Types of Matrices

- **Row matrix:** A matrix with exactly one row, e.g. $\begin{bmatrix} 1 & 2 & 3 \end{bmatrix}$ is a 1×3 row matrix.
- **Column matrix:** A matrix with exactly one column, e.g. $\begin{bmatrix} 4 \\ 5 \end{bmatrix}$ is a 2×1 column matrix.
- **Square matrix:** A matrix where $m = n$ (same number of rows and columns).
- **Zero matrix (O):** A matrix in which every entry is 0. For example, $O_{2 \times 2} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$.
- **Identity matrix (I):** A square matrix with 1s on the main diagonal and 0s everywhere else. For any square matrix A of the same size, $AI = IA = A$.

$$I_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}, \quad I_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

- **Diagonal matrix:** A square matrix where all off-diagonal entries are 0, e.g. $\begin{bmatrix} 3 & 0 \\ 0 & -7 \end{bmatrix}$.
- **Symmetric matrix:** A square matrix A where $A = A^T$ (equal to its transpose), e.g. $\begin{bmatrix} 1 & 4 \\ 4 & 2 \end{bmatrix}$.

Matrix Operations

1. Matrix Addition and Subtraction

Two matrices can be added (or subtracted) only if they have the *same dimensions*. The operation is performed entry by entry:

$$A + B = \begin{bmatrix} a_{11} + b_{11} & a_{12} + b_{12} \\ a_{21} + b_{21} & a_{22} + b_{22} \end{bmatrix}$$

2. Scalar Multiplication

To multiply a matrix by a scalar k , multiply every entry by k :

$$kA = \begin{bmatrix} ka_{11} & ka_{12} \\ ka_{21} & ka_{22} \end{bmatrix}$$

3. Matrix Multiplication

The product AB is defined only when the number of *columns* of A equals the number of *rows* of B . If A is $m \times n$ and B is $n \times p$, then AB is $m \times p$.

The entry in row i , column j of AB is the dot product of row i of A with column j of B :

$$(AB)_{ij} = \sum_{k=1}^n a_{ik} b_{kj}$$

Remark 2.1 (Key Properties of Matrix Multiplication)

- **Not commutative:** In general, $AB \neq BA$ (even when both products are defined).
- **Associative:** $(AB)C = A(BC)$.
- **Distributive:** $A(B + C) = AB + AC$ and $(A + B)C = AC + BC$.
- **Identity:** $AI = IA = A$.
- **Zero:** $AO = OA = O$ (where O is the zero matrix of appropriate size).

4. Transpose

The **transpose** of an $m \times n$ matrix A , written A^T , is the $n \times m$ matrix obtained by swapping its rows and columns: the entry in row i , column j of A^T is a_{ji} .

Key properties of the transpose:

- $(A^T)^T = A$
- $(A + B)^T = A^T + B^T$
- $(kA)^T = kA^T$
- $(AB)^T = B^T A^T$ (note the reversal of order)

Problem 2.1: Matrix Algebra — Basic Operations

Let $A = \begin{bmatrix} 2 & -1 \\ 3 & 4 \end{bmatrix}$, $B = \begin{bmatrix} 0 & 5 \\ -2 & 1 \end{bmatrix}$, and $k = 3$.

- Calculate $A + B$.
- Calculate kA .
- Calculate AB .
- Calculate BA .
- Is $AB = BA$? What does this tell you about matrix multiplication?

Solution 2.1

(a)

$$A + B = \begin{bmatrix} 2+0 & -1+5 \\ 3+(-2) & 4+1 \end{bmatrix} = \begin{bmatrix} 2 & 4 \\ 1 & 5 \end{bmatrix}$$

(b)

$$kA = 3 \begin{bmatrix} 2 & -1 \\ 3 & 4 \end{bmatrix} = \begin{bmatrix} 6 & -3 \\ 9 & 12 \end{bmatrix}$$

(c)

$$AB = \begin{bmatrix} 2 & -1 \\ 3 & 4 \end{bmatrix} \begin{bmatrix} 0 & 5 \\ -2 & 1 \end{bmatrix}$$

Row 1 of A · columns of B :

$$(2)(0) + (-1)(-2) = 2, \quad (2)(5) + (-1)(1) = 9$$

Row 2 of A · columns of B :

$$(3)(0) + (4)(-2) = -8, \quad (3)(5) + (4)(1) = 19$$

$$AB = \begin{bmatrix} 2 & 9 \\ -8 & 19 \end{bmatrix}$$

(d)

$$BA = \begin{bmatrix} 0 & 5 \\ -2 & 1 \end{bmatrix} \begin{bmatrix} 2 & -1 \\ 3 & 4 \end{bmatrix}$$

$$(0)(2) + (5)(3) = 15, \quad (0)(-1) + (5)(4) = 20$$

$$(-2)(2) + (1)(3) = -1, \quad (-2)(-1) + (1)(4) = 6$$

$$BA = \begin{bmatrix} 15 & 20 \\ -1 & 6 \end{bmatrix}$$

(e)

$$AB = \begin{bmatrix} 2 & 9 \\ -8 & 19 \end{bmatrix} \neq \begin{bmatrix} 15 & 20 \\ -1 & 6 \end{bmatrix} = BA$$

Matrix multiplication is **not commutative**. The order in which you multiply matrices matters. This is one of the most important differences between matrix algebra and ordinary number algebra.

Takeaways 2.1

- **Dimension Compatibility:** Always check dimensions before multiplying — an $m \times n$ matrix can only multiply an $n \times p$ matrix.
- **Non-Commutativity:** $AB \neq BA$ in general. Never assume you can swap the order of matrix multiplication.
- **Entry-by-Entry vs. Row-by-Column:** Addition is entry-by-entry (same dimensions required). Multiplication uses the row-by-column dot product rule (inner dimensions must match).

Problem 2.2: Matrix Algebra — Transpose

Let $P = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$ and $Q = \begin{bmatrix} 7 & 8 \\ 9 & 10 \\ 11 & 12 \end{bmatrix}$.

- (a) Find P^T and state its dimensions.
 (b) Calculate PQ and then find $(PQ)^T$.
 (c) Calculate $Q^T P^T$ and verify that $(PQ)^T = Q^T P^T$.

Solution 2.2

(a)

$$P^T = \begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix}$$

P is 2×3 , so P^T is 3×2 .

(b)

P is 2×3 and Q is 3×2 , so PQ is 2×2 .

$$PQ = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix} \begin{bmatrix} 7 & 8 \\ 9 & 10 \\ 11 & 12 \end{bmatrix}$$

$$(1)(7) + (2)(9) + (3)(11) = 58, \quad (1)(8) + (2)(10) + (3)(12) = 64$$

$$(4)(7) + (5)(9) + (6)(11) = 139, \quad (4)(8) + (5)(10) + (6)(12) = 154$$

$$PQ = \begin{bmatrix} 58 & 64 \\ 139 & 154 \end{bmatrix}, \quad (PQ)^T = \begin{bmatrix} 58 & 139 \\ 64 & 154 \end{bmatrix}$$

(c)

$$Q^T = \begin{bmatrix} 7 & 9 & 11 \\ 8 & 10 & 12 \end{bmatrix} \quad (2 \times 3), \quad P^T = \begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix} \quad (3 \times 2)$$

$$Q^T P^T = \begin{bmatrix} 7 & 9 & 11 \\ 8 & 10 & 12 \end{bmatrix} \begin{bmatrix} 1 & 4 \\ 2 & 5 \\ 3 & 6 \end{bmatrix}$$

$$(7)(1) + (9)(2) + (11)(3) = 58, \quad (7)(4) + (9)(5) + (11)(6) = 139$$

$$(8)(1) + (10)(2) + (12)(3) = 64, \quad (8)(4) + (10)(5) + (12)(6) = 154$$

$$Q^T P^T = \begin{bmatrix} 58 & 139 \\ 64 & 154 \end{bmatrix}$$

$$\text{Confirmed: } (PQ)^T = Q^T P^T = \begin{bmatrix} 58 & 139 \\ 64 & 154 \end{bmatrix}. \quad \checkmark$$

Takeaways 2.2

- **Transpose Reverses Dimensions:** If A is $m \times n$, then A^T is $n \times m$.
- **Shoe–Sock Rule:** The transpose of a product reverses the order: $(AB)^T = B^T A^T$. Think of it like putting on socks then shoes — to reverse, you remove shoes first, then socks.

2.2 Inverses and Determinants for 2×2 Matrices

In standard algebra, the inverse of a number x (where $x \neq 0$) is $\frac{1}{x}$, such that $x \times \frac{1}{x} = 1$. Similarly, in matrix algebra, the **inverse** of a square matrix A , denoted A^{-1} , is the matrix that multiplies with A to yield the identity matrix I .

$$AA^{-1} = A^{-1}A = I$$

The Determinant

The **determinant** of a matrix is a special scalar value that tells us whether the matrix has an inverse. For a 2×2 matrix:

$$A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$$

The determinant is denoted by $\det(A)$ or $|A|$ and is calculated as:

$$\det(A) = ad - bc$$

The Inverse Formula

If $\det(A) \neq 0$, the inverse of the 2×2 matrix A exists and is given by:

$$A^{-1} = \frac{1}{\det(A)} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$$

Notice the pattern for finding the inverse of a 2×2 matrix:

- **Swap** the elements on the main diagonal (a and d).
- **Negate** the elements on the other diagonal (b and c).
- **Multiply** the resulting matrix by the scalar $\frac{1}{\det(A)}$.

Remark 2.2 (Singular Matrices)

If $\det(A) = 0$, then $\frac{1}{\det(A)}$ is undefined. In this case, the matrix A **does not have an inverse** and is called a **singular matrix**. If $\det(A) \neq 0$, the matrix is called **non-singular**.

Problem 2.3: Calculating Inverses and Determinants

Consider the following two matrices:

$$M = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix}, \quad N = \begin{bmatrix} 6 & -2 \\ -9 & 3 \end{bmatrix}$$

- Calculate $\det(M)$. Hence, find M^{-1} .
- Verify that $MM^{-1} = I$.
- Determine whether matrix N has an inverse.

Solution 2.3**(a)**First, find the determinant of M :

$$\det(M) = (4)(3) - (1)(2) = 12 - 2 = 10.$$

Since $\det(M) = 10 \neq 0$, the inverse exists.Apply the 2×2 inverse formula (swap main diagonal, negate the other):

$$M^{-1} = \frac{1}{10} \begin{bmatrix} 3 & -1 \\ -2 & 4 \end{bmatrix}.$$

*(Note: Leaving the scalar $1/10$ at the front is usually preferred unless decimals/fractions are specifically required for further calculation.)***(b)**To verify, calculate the product MM^{-1} :

$$MM^{-1} = \begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix} \left(\frac{1}{10} \begin{bmatrix} 3 & -1 \\ -2 & 4 \end{bmatrix} \right)$$

Because scalar multiplication can be pulled to the front, we calculate the matrix product first:

$$\begin{aligned} MM^{-1} &= \frac{1}{10} \left(\begin{bmatrix} 4 & 1 \\ 2 & 3 \end{bmatrix} \begin{bmatrix} 3 & -1 \\ -2 & 4 \end{bmatrix} \right) \\ &= \frac{1}{10} \begin{bmatrix} (4)(3) + (1)(-2) & (4)(-1) + (1)(4) \\ (2)(3) + (3)(-2) & (2)(-1) + (3)(4) \end{bmatrix} \\ &= \frac{1}{10} \begin{bmatrix} 12 - 2 & -4 + 4 \\ 6 - 6 & -2 + 12 \end{bmatrix} = \frac{1}{10} \begin{bmatrix} 10 & 0 \\ 0 & 10 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I. \quad \checkmark \end{aligned}$$

(c)First, find the determinant of N :

$$\det(N) = (6)(3) - (-2)(-9) = 18 - 18 = 0.$$

Because $\det(N) = 0$, matrix N is a **singular matrix**. It does not have an inverse.**Takeaways 2.3**

- **Determinant First:** Always calculate the determinant before attempting to find the inverse. If it is zero, you can state that the matrix is singular and stop immediately.
- **Scalar Extraction:** When multiplying a matrix by its inverse to check your answer, it is much easier to leave the scalar fraction (like $\frac{1}{10}$) outside the matrix multiplication until the very end.
- **Algebraic vs. Geometric:** A singular matrix ($\det = 0$) cannot be inverted algebraically (as this would require division by zero). Geometrically, this means the transformation squashes 2D space down into a 1D line or a point, losing information that cannot be uniquely reconstructed.

2.3 Simultaneous Linear Equations with Two Variables

Before using matrices to solve equations, it is important to review what a system of simultaneous linear equations actually represents. A linear equation in two variables, such as $ax + by = c$, geometrically represents a straight line on a Cartesian plane.

When we solve two simultaneous linear equations, we are looking for the (x, y) coordinates that satisfy both equations at the same time. Geometrically, this means finding the point(s) where the two lines intersect.

Three Possible Outcomes

For any system of two linear equations:

$$\begin{cases} a_1x + b_1y = c_1 \\ a_2x + b_2y = c_2 \end{cases}$$

There are three possible geometric configurations and algebraic outcomes:

- **Unique Solution:** The lines have different gradients and intersect at exactly **one point**.
- **No Solution:** The lines have the same gradient but different y -intercepts. They are **parallel** and never intersect. Algebraically, solving them yields a contradiction (e.g., $0 = 5$).
- **Infinitely Many Solutions:** The lines have the same gradient and the same y -intercept. They are **coincident** (the exact same line). Algebraically, solving them yields a tautology (e.g., $0 = 0$).

Problem 2.4: Algebraic and Geometric Interpretations

For each of the following systems of equations, solve algebraically using either substitution or elimination, and describe the geometric relationship between the two lines.

(a)

$$2x + 3y = 8$$

$$5x - y = 3$$

(b)

$$x + 2y = 4$$

$$2x + 4y = 10$$

Solution 2.4**(a)**

Let the equations be:

$$2x + 3y = 8 \quad \text{--- (1)}$$

$$5x - y = 3 \quad \text{--- (2)}$$

Using substitution, rearrange equation (2) to make y the subject:

$$y = 5x - 3$$

Substitute this into equation (1):

$$2x + 3(5x - 3) = 8$$

$$2x + 15x - 9 = 8$$

$$17x = 17$$

$$x = 1$$

Substitute $x = 1$ back into the rearranged equation (2):

$$y = 5(1) - 3 = 2$$

Algebraic Solution: $x = 1, y = 2$.**Geometric Relationship:** The two lines have a unique solution. They intersect at the single point $(1, 2)$.**(b)**

Let the equations be:

$$x + 2y = 4 \quad \text{--- (1)}$$

$$2x + 4y = 10 \quad \text{--- (2)}$$

Using elimination, multiply equation (1) by 2:

$$2x + 4y = 8 \quad \text{--- (3)}$$

Subtract equation (3) from equation (2):

$$(2x + 4y) - (2x + 4y) = 10 - 8$$

$$0 = 2$$

This is a mathematical contradiction (0 can never equal 2).**Algebraic Solution:** No solution.**Geometric Relationship:** The two equations represent distinct parallel lines. Because they are parallel, they never intersect.**Takeaways 2.4**

- **Algebra meets Geometry:** Solving simultaneous equations is identical to finding the intersection points of lines on a graph.
- **Contradictions reveal Parallel Lines:** If your algebraic steps lead to an impossible statement like $0 = 2$, it is not a mistake. It is the mathematical way of stating the lines are parallel.

2.4 Using Matrix Algebra for Systems of Linear Equations in Two Variables

Any system of linear equations can be translated into a single matrix equation. This allows us to use the power of matrix inverses to solve for the variables directly, bypassing manual substitution or elimination.

Forming the Matrix Equation

Consider the general 2×2 system:

$$\begin{cases} ax + by = e \\ cx + dy = f \end{cases}$$

By using the rules of matrix multiplication, we can separate the coefficients, the variables, and the constants into three matrices:

$$\begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} e \\ f \end{bmatrix}$$

This is often written compactly as $AX = B$, where:

- A is the **coefficient matrix**.
- X is the **variable matrix** (a column vector).
- B is the **constant matrix** (a column vector).

Solving the Matrix Equation

To isolate the variable matrix X , we cannot “divide” by matrix A . Instead, we multiply both sides of the equation by the inverse matrix, A^{-1} .

Because matrix multiplication is not commutative, we must multiply from the **left** on both sides:

$$\begin{aligned} AX &= B \\ A^{-1}(AX) &= A^{-1}B \\ (A^{-1}A)X &= A^{-1}B \\ IX &= A^{-1}B \\ X &= A^{-1}B \end{aligned}$$

If $\det(A) = 0$, then A^{-1} does not exist. This algebraically confirms the situations discussed in the previous topic: the system has either no unique solution (parallel lines) or infinitely many solutions (coincident lines).

Problem 2.5: Matrix Solution to a 2×2 System

Solve the following system of equations using matrix algebra:

$$\begin{aligned} 2x + 3y &= 8 \\ 5x - y &= 3 \end{aligned}$$

(Note: This is the same system solved algebraically in Topic 3.)

Solution 2.5**Step 1: Write the system as a matrix equation.**

Ensure the equations are in standard form (variables on the left, constants on the right). They are.

$$AX = B \implies \begin{bmatrix} 2 & 3 \\ 5 & -1 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 8 \\ 3 \end{bmatrix}$$

Step 2: Find the determinant of A .

$$\det(A) = (2)(-1) - (3)(5) = -2 - 15 = -17.$$

Since $\det(A) \neq 0$, a unique solution exists.**Step 3: Find the inverse matrix, A^{-1} .**

$$A^{-1} = \frac{1}{-17} \begin{bmatrix} -1 & -3 \\ -5 & 2 \end{bmatrix} = -\frac{1}{17} \begin{bmatrix} -1 & -3 \\ -5 & 2 \end{bmatrix}$$

Step 4: Solve for X using $X = A^{-1}B$.

$$X = -\frac{1}{17} \begin{bmatrix} -1 & -3 \\ -5 & 2 \end{bmatrix} \begin{bmatrix} 8 \\ 3 \end{bmatrix}$$

Calculate the matrix product first:

$$X = -\frac{1}{17} \begin{bmatrix} (-1)(8) + (-3)(3) \\ (-5)(8) + (2)(3) \end{bmatrix}$$

$$X = -\frac{1}{17} \begin{bmatrix} -8 - 9 \\ -40 + 6 \end{bmatrix} = -\frac{1}{17} \begin{bmatrix} -17 \\ -34 \end{bmatrix}$$

Finally, multiply the scalar into the matrix:

$$X = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$$

Conclusion:Since $X = \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$, the solution is $x = 1, y = 2$.This precisely matches the geometric point of intersection $(1, 2)$ found previously!**Takeaways 2.5**

- **Left-Multiplication is Critical:** You must write $X = A^{-1}B$. Writing $X = BA^{-1}$ is mathematically incorrect because the inner dimensions for multiplication would not match (a 2×1 matrix cannot be multiplied by a 2×2 matrix in that order).
- **Scalar Timing:** As seen in the example, delay multiplying the scalar fraction (like $-\frac{1}{17}$) until the very last step. This avoids messy fraction arithmetic during the matrix multiplication.
- **Determinant Check:** The determinant provides an immediate check. If $\det(A) = 0$, you can stop and state that no unique geometric intersection exists.

2.5 Inverses and Determinants for $n \times n$ MatricesWhile the inverse of a 2×2 matrix has a simple formula, extending this to 3×3 matrices (and larger) requires a more systematic process using **minors** and **cofactors**.**Minors and Cofactors**For any element a_{ij} in a square matrix A :

- The **minor**, M_{ij} , is the determinant of the smaller matrix that remains after deleting row i and column j .

- The **cofactor**, C_{ij} , is the minor multiplied by a positive or negative sign, based on its position:

$$C_{ij} = (-1)^{i+j} M_{ij}$$

This creates a “checkerboard” pattern of signs. For a 3×3 matrix:

$$\begin{bmatrix} + & - & + \\ - & + & - \\ + & - & + \end{bmatrix}$$

Evaluating a 3×3 Determinant

The determinant of a 3×3 matrix can be evaluated by **cofactor expansion** along any row or column. You multiply each element in that row (or column) by its corresponding cofactor and add them up.

Expanding along the first row:

$$\det(A) = a_{11}C_{11} + a_{12}C_{12} + a_{13}C_{13}$$

Using the 2×2 determinant formula for the minors, this expands to:

$$\det(A) = a_{11}(a_{22}a_{33} - a_{23}a_{32}) - a_{12}(a_{21}a_{33} - a_{23}a_{31}) + a_{13}(a_{21}a_{32} - a_{22}a_{31})$$

The Inverse of an $n \times n$ Matrix

The formula for the inverse of any non-singular $n \times n$ matrix is:

$$A^{-1} = \frac{1}{\det(A)} \text{adj}(A)$$

Where $\text{adj}(A)$, called the **adjugate matrix**, is the *transpose* of the matrix of cofactors.

Problem 2.6: Inverting a 3×3 Matrix

Let $A = \begin{bmatrix} 1 & 2 & -1 \\ 0 & 3 & 1 \\ 2 & 0 & -2 \end{bmatrix}$.

- Evaluate $\det(A)$ by expanding along the first row.
- Find the matrix of cofactors, C .
- Find the adjugate matrix, $\text{adj}(A)$.
- State the inverse matrix, A^{-1} .

Solution 2.6**(a) Determinant**

Expanding along the first row:

$$\begin{aligned}\det(A) &= 1 \begin{vmatrix} 3 & 1 \\ 0 & -2 \end{vmatrix} - 2 \begin{vmatrix} 0 & 1 \\ 2 & -2 \end{vmatrix} + (-1) \begin{vmatrix} 0 & 3 \\ 2 & 0 \end{vmatrix} \\ &= 1((3)(-2) - (1)(0)) - 2((0)(-2) - (1)(2)) - 1((0)(0) - (3)(2)) \\ &= 1(-6) - 2(-2) - 1(-6) \\ &= -6 + 4 + 6 = 4\end{aligned}$$

Since $\det(A) = 4 \neq 0$, the inverse exists.

(b) Matrix of Cofactors

We calculate the cofactor for all 9 elements. Remember to apply the checkerboard signs!

$$\begin{aligned}C_{11} &= +((3)(-2) - (1)(0)) = -6 \\ C_{12} &= -((0)(-2) - (1)(2)) = -(-2) = 2 \\ C_{13} &= +((0)(0) - (3)(2)) = -6 \\ C_{21} &= -((2)(-2) - (-1)(0)) = -(-4) = 4 \\ C_{22} &= +((1)(-2) - (-1)(2)) = -2 + 2 = 0 \\ C_{23} &= -((1)(0) - (2)(2)) = -(-4) = 4 \\ C_{31} &= +((2)(1) - (-1)(3)) = 2 + 3 = 5 \\ C_{32} &= -((1)(1) - (-1)(0)) = -(1) = -1 \\ C_{33} &= +((1)(3) - (2)(0)) = 3\end{aligned}$$

The cofactor matrix is:

$$C = \begin{bmatrix} -6 & 2 & -6 \\ 4 & 0 & 4 \\ 5 & -1 & 3 \end{bmatrix}$$

(c) Adjugate Matrix

The adjugate is the transpose of the cofactor matrix (C^T):

$$\text{adj}(A) = \begin{bmatrix} -6 & 4 & 5 \\ 2 & 0 & -1 \\ -6 & 4 & 3 \end{bmatrix}$$

(d) Inverse Matrix

Using the formula $A^{-1} = \frac{1}{\det(A)} \text{adj}(A)$:

$$A^{-1} = \frac{1}{4} \begin{bmatrix} -6 & 4 & 5 \\ 2 & 0 & -1 \\ -6 & 4 & 3 \end{bmatrix}$$

Takeaways 2.6

- **Row Choice for Determinants:** You can expand along *any* row or column to find the determinant. For speed, always choose the row or column containing the most zeros.
- **Don't Forget the Transpose:** A common error is writing the cofactor matrix directly into the inverse formula. You must take the transpose of the cofactor matrix to get the adjugate.
- **Sign Errors:** Calculating a 3×3 inverse manually involves many small multiplications and subtractions. Write out your working explicitly to avoid dropping minus signs.

2.6 Simultaneous Linear Equations with More Than Two Variables

When we move from two variables (x, y) to three variables (x, y, z) , the geometric interpretation expands from two-dimensional space to three-dimensional space.

A linear equation in two variables $(ax + by = c)$ represents a **line**.

A linear equation in three variables $(ax + by + cz = d)$ represents a **flat plane** in 3D space.

Geometric Interpretations in 3D

When solving a system of three linear equations in three variables, we are looking for the point(s) in 3D space where all three planes intersect. There are three main categories of outcomes:

- **Unique Solution:** The three planes meet at a single, distinct point.
- **Infinitely Many Solutions:** The planes intersect along a common line (like the pages of a book meeting at the spine), or all three equations represent the exact same plane.
- **No Solution:** There is no single point that belongs to all three planes. For example, two planes might be parallel, or they might form a triangular prism (like a tent) where they intersect in pairs but never all three together.

Setting Up Systems

To find a unique solution for n unknown variables, you generally need n independent linear equations. A common application is translating real-world scenarios into a 3×3 system of equations.

Problem 2.7: Translating a Word Problem into a System

A local bakery sells three types of cakes: chocolate, vanilla, and strawberry. The bakery records its sales and revenue for three consecutive days:

- **Monday:** 2 chocolate, 3 vanilla, and 1 strawberry cake were sold for a total of \$115.
- **Tuesday:** 1 chocolate, 2 vanilla, and 2 strawberry cakes were sold for a total of \$110.
- **Wednesday:** 3 chocolate, 1 vanilla, and 3 strawberry cakes were sold for a total of \$165.

Assuming the price of each cake type remains constant, define appropriate variables and set up a system of three simultaneous linear equations that models this situation.

Solution 2.7**Step 1: Define the variables.**

Let x = the price of a chocolate cake (in \$).

Let y = the price of a vanilla cake (in \$).

Let z = the price of a strawberry cake (in \$).

Step 2: Translate each day's sales into an equation.

Monday's sales (2 choc, 3 van, 1 straw yield \$115):

$$2x + 3y + z = 115$$

Tuesday's sales (1 choc, 2 van, 2 straw yield \$110):

$$x + 2y + 2z = 110$$

Wednesday's sales (3 choc, 1 van, 3 straw yield \$165):

$$3x + y + 3z = 165$$

Step 3: State the final system.

The system of simultaneous linear equations is:

$$2x + 3y + z = 115$$

$$x + 2y + 2z = 110$$

$$3x + y + 3z = 165$$

(Note: We will solve this exact system using matrices in the next topic!)

Takeaways 2.7

- **Geometry of 3 Variables:** Always visualize 3×3 systems as three flat planes in 3D space. The algebraic solution (x, y, z) is their exact point of intersection.
- **Information Requirement:** Solving for 3 unknowns uniquely requires 3 distinct pieces of information (equations). If you only have two equations, you can at best find a line of intersection (infinitely many solutions), not a unique point.
- **Careful Translation:** When reading word problems, ensure the coefficients (quantities) align correctly with their corresponding variables (prices) in each equation.

2.7 Using Matrix Algebra for Systems of Linear Equations in More Than Two Variables

The elegance of matrix algebra is that the fundamental equation used to solve a system, $X = A^{-1}B$, remains exactly the same regardless of whether you are solving a 2×2 system, a 3×3 system, or a 100×100 system.

The only difference is the computational effort required to find the inverse matrix, A^{-1} .

Problem 2.8: Matrix Solution to a 3×3 System

Recall the bakery word problem from the previous topic, which gave us the following system:

$$2x + 3y + z = 115$$

$$x + 2y + 2z = 110$$

$$3x + y + 3z = 165$$

Use matrix algebra to solve for x , y , and z , finding the exact price of each cake.

Solution 2.8

Step 1: Write the system in the form $AX = B$.

$$\begin{bmatrix} 2 & 3 & 1 \\ 1 & 2 & 2 \\ 3 & 1 & 3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 115 \\ 110 \\ 165 \end{bmatrix}$$

Where A is the 3×3 coefficient matrix, and B is the constants matrix.

Step 2: Find the determinant of A .

Expanding along the first row:

$$\begin{aligned} \det(A) &= 2 \begin{vmatrix} 2 & 2 \\ 1 & 3 \end{vmatrix} - 3 \begin{vmatrix} 1 & 2 \\ 3 & 3 \end{vmatrix} + 1 \begin{vmatrix} 1 & 2 \\ 3 & 1 \end{vmatrix} \\ &= 2(6 - 2) - 3(3 - 6) + 1(1 - 6) \\ &= 2(4) - 3(-3) + 1(-5) \\ &= 8 + 9 - 5 = 12 \end{aligned}$$

Since $\det(A) = 12 \neq 0$, a unique solution exists.

Step 3: Find the matrix of cofactors, C .

Calculating the 9 cofactors (with the checkerboard sign pattern):

$$\begin{array}{lll} C_{11} = +(6 - 2) = 4, & C_{12} = -(3 - 6) = 3, & C_{13} = +(1 - 6) = -5 \\ C_{21} = -(9 - 1) = -8, & C_{22} = +(6 - 3) = 3, & C_{23} = -(2 - 9) = 7 \\ C_{31} = +(6 - 2) = 4, & C_{32} = -(4 - 1) = -3, & C_{33} = +(4 - 3) = 1 \end{array}$$

So, the cofactor matrix is $C = \begin{bmatrix} 4 & 3 & -5 \\ -8 & 3 & 7 \\ 4 & -3 & 1 \end{bmatrix}$.

Step 4: Find the adjugate and inverse of A .

Transpose C to get the adjugate:

$$\text{adj}(A) = C^T = \begin{bmatrix} 4 & -8 & 4 \\ 3 & 3 & -3 \\ -5 & 7 & 1 \end{bmatrix}$$

The inverse is $A^{-1} = \frac{1}{\det(A)} \text{adj}(A) = \frac{1}{12} \begin{bmatrix} 4 & -8 & 4 \\ 3 & 3 & -3 \\ -5 & 7 & 1 \end{bmatrix}$.

Step 5: Solve for X using $X = A^{-1}B$.

$$X = \frac{1}{12} \begin{bmatrix} 4 & -8 & 4 \\ 3 & 3 & -3 \\ -5 & 7 & 1 \end{bmatrix} \begin{bmatrix} 115 \\ 110 \\ 165 \end{bmatrix}$$

Calculate the matrix product first:

$$\begin{aligned} X &= \frac{1}{12} \begin{bmatrix} 4(115) + (-8)(110) + 4(165) \\ 3(115) + 3(110) + (-3)(165) \\ -5(115) + 7(110) + 1(165) \end{bmatrix} \\ &= \frac{1}{12} \begin{bmatrix} 460 - 880 + 660 \\ 345 + 330 - 495 \\ -575 + 770 + 165 \end{bmatrix} = \frac{1}{12} \begin{bmatrix} 240 \\ 180 \\ 360 \end{bmatrix} = \begin{bmatrix} 20 \\ 15 \\ 30 \end{bmatrix} \end{aligned}$$

Conclusion:

The variables are $x = 20, y = 15, z = 30$.

The prices are \$20 for a chocolate cake, \$15 for a vanilla cake, and \$30 for a strawberry cake.

Takeaways 2.8

- **Universal Logic:** The framework $X = A^{-1}B$ scales to any number of dimensions. Only the mechanics of finding A^{-1} change.
- **Computational Load:** As systems get larger (e.g., 4×4), finding inverses by hand becomes highly prone to arithmetic errors. This is why computers and graphics cards are used to solve these systems in real life!
- **Checking your work:** You can easily verify the final solution by substituting the values back into any of the original equations (e.g., $2(20)+3(15)+30 = 40+45+30 = 115$, which matches Monday's total).

2.8 Using Augmented Matrices for Systems of Equations

Finding the inverse of a 3×3 or larger matrix is computationally heavy. Furthermore, if a system has no solutions or infinitely many solutions, the inverse method fails because $\det(A) = 0$.

A more robust and efficient method is **Gaussian elimination**, which uses **augmented matrices** and **elementary row operations**.

The Augmented Matrix

Instead of writing three separate matrices (A , X , and B), we combine the coefficient matrix A and the constant matrix B into a single matrix, separated by a vertical line. For a 3×3 system, it looks like this:

$$[A | B] = \left[\begin{array}{ccc|c} a_{11} & a_{12} & a_{13} & b_1 \\ a_{21} & a_{22} & a_{23} & b_2 \\ a_{31} & a_{32} & a_{33} & b_3 \end{array} \right]$$

Elementary Row Operations

Because each row represents an algebraic equation, we can perform operations on the rows without changing the solution to the system. There are three valid **elementary row operations**:

1. **Swap:** Interchange any two rows (e.g., $R_1 \leftrightarrow R_2$).
2. **Scale:** Multiply an entire row by a non-zero scalar (e.g., $R_2 \rightarrow \frac{1}{2}R_2$).
3. **Add/Subtract:** Add a multiple of one row to another row (e.g., $R_3 \rightarrow R_3 - 2R_1$).

Row Echelon Form (REF)

The goal of Gaussian elimination is to use these operations to transform the augmented matrix into **Row Echelon Form**. In REF, the bottom-left corner of the matrix is filled with zeros, creating a "staircase" pattern:

$$\left[\begin{array}{ccc|c} 1 & * & * & * \\ 0 & 1 & * & * \\ 0 & 0 & 1 & * \end{array} \right]$$

Once in this form, the system can be easily solved from the bottom up using **back-substitution**.

Problem 2.9: Gaussian Elimination

Solve the following system of linear equations using an augmented matrix and row reduction:

$$\begin{aligned}x + y + z &= 6 \\2x - y + 2z &= 6 \\x + 2y - z &= 2\end{aligned}$$

Solution 2.9

Step 1: Write the augmented matrix.

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 2 & -1 & 2 & 6 \\ 1 & 2 & -1 & 2 \end{array} \right]$$

Step 2: Eliminate the x -coefficients in Rows 2 and 3.

To get a zero in R_2 , column 1, we subtract $2 \times R_1$ from R_2 ($R_2 \rightarrow R_2 - 2R_1$):

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & -3 & 0 & -6 \\ 1 & 2 & -1 & 2 \end{array} \right]$$

To get a zero in R_3 , column 1, we subtract R_1 from R_3 ($R_3 \rightarrow R_3 - R_1$):

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & -3 & 0 & -6 \\ 0 & 1 & -2 & -4 \end{array} \right]$$

Step 3: Simplify and eliminate the y -coefficient in Row 3.

Let's simplify R_2 by dividing by -3 ($R_2 \rightarrow -\frac{1}{3}R_2$):

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & 0 & 2 \\ 0 & 1 & -2 & -4 \end{array} \right]$$

Now, eliminate the y -coefficient in R_3 by subtracting R_2 ($R_3 \rightarrow R_3 - R_2$):

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & -2 & -6 \end{array} \right]$$

Step 4: Finalize Row Echelon Form and Back-Substitute.

Divide R_3 by -2 ($R_3 \rightarrow -\frac{1}{2}R_3$):

$$\left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & 0 & 2 \\ 0 & 0 & 1 & 3 \end{array} \right]$$

Translate the matrix back into equations:

$$\begin{aligned}x + y + z &= 6 & \text{--- (1)} \\ y &= 2 & \text{--- (2)} \\ z &= 3 & \text{--- (3)}\end{aligned}$$

We already have $y = 2$ and $z = 3$. Substitute these back into equation (1):

$$\begin{aligned}x + (2) + (3) &= 6 \\ x + 5 &= 6 \\ x &= 1\end{aligned}$$

The unique solution is $x = 1, y = 2, z = 3$.

Takeaways 2.9

- **Computational Efficiency:** Gaussian elimination requires far fewer calculations than finding a 3×3 inverse, making it the preferred method for solving systems by hand.
- **Detecting Anomalies:** If a system has no solution, row reduction will eventually produce an impossible row like $\begin{bmatrix} 0 & 0 & 0 & | & 5 \end{bmatrix}$, which algebraically means $0 = 5$.
- **Infinite Solutions:** If a system has infinite solutions, row reduction will produce a redundant row like $\begin{bmatrix} 0 & 0 & 0 & | & 0 \end{bmatrix}$, indicating one equation was just a multiple of another.

2.9 Dominance Matrices

Matrices are not only used for solving algebraic equations; they are incredibly powerful tools in discrete mathematics and network theory. A **dominance matrix** is used to represent and analyze directed relationships, such as the outcomes of a round-robin sports tournament or a “pecking order” in a group of animals.

Constructing the Dominance Matrix

If a group of n competitors play each other exactly once (no draws allowed), the results can be stored in an $n \times n$ matrix D :

- $d_{ij} = 1$ if competitor i defeated competitor j .
- $d_{ij} = 0$ if competitor i lost to competitor j .

The main diagonal is always entirely zeros, as a competitor cannot play themselves.

Direct and Indirect Wins

- **One-step dominance (Direct wins):** The sum of the elements in row i of matrix D gives the total number of matches won by competitor i . However, this often results in ties.
- **Two-step dominance (Indirect wins):** If Team A beats Team B, and Team B beats Team C, Team A has a two-step dominance over Team C. Mathematically, this is found by squaring the matrix to get D^2 . This rewards teams for beating *strong* opponents.
- **Total Dominance Score:** To break ties, we calculate a power score matrix $T = D + D^2$. The row sums of T provide a much more robust ranking system.

Problem 2.10: Tournament Rankings

Four teams (A, B, C, and D) compete in a round-robin tournament. The results are as follows:

- Team A defeated Teams B and C.
 - Team B defeated Teams C and D.
 - Team C defeated Team D.
 - Team D defeated Team A.
- (a) Construct the 4×4 dominance matrix D . Calculate the row sums to find the direct rankings and identify any ties.
- (b) Calculate D^2 , the two-step dominance matrix.
- (c) Calculate the total dominance matrix $T = D + D^2$, and use its row sums to determine the final, tie-broken ranking of the four teams.

Solution 2.10**(a) Matrix D and Direct Rankings**

Following the rules, we place a 1 in row i , column j if the row team beat the column team.

$$D = \begin{bmatrix} 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \end{bmatrix}$$

Row Sums (Direct Wins):

- Team A sum = 2
- Team B sum = 2
- Team C sum = 1
- Team D sum = 1

We have a tie for 1st place (A and B) and a tie for 3rd place (C and D).

(b) Matrix D^2 (Indirect Wins)

Multiply D by itself. For example, to find Row 1, Col 4 of D^2 , we calculate $(0)(0) + (1)(1) + (1)(1) + (0)(0) = 2$. This means Team A has two indirect pathways to beat Team D (via B and via C).

$$D^2 = D \times D = \begin{bmatrix} 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 & 2 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \end{bmatrix}$$

(c) Total Dominance and Final Ranking

Calculate $T = D + D^2$ by adding the corresponding elements:

$$T = \begin{bmatrix} 0 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \end{bmatrix} + \begin{bmatrix} 0 & 0 & 1 & 2 \\ 1 & 0 & 0 & 1 \\ 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 2 & 2 \\ 1 & 0 & 1 & 2 \\ 1 & 0 & 0 & 1 \\ 1 & 1 & 1 & 0 \end{bmatrix}$$

Row Sums of T :

- Team A = $0 + 1 + 2 + 2 = 5$
- Team B = $1 + 0 + 1 + 2 = 4$
- Team D = $1 + 1 + 1 + 0 = 3$
- Team C = $1 + 0 + 0 + 1 = 2$

Final Ranking: 1st: Team A, 2nd: Team B, 3rd: Team D, 4th: Team C.

The ties have been successfully broken!

Takeaways 2.10

- **Quality of Wins:** Team A and B both had 2 direct wins. However, Team A beat Team B (a strong team), giving A more indirect pathways to dominance. The $D + D^2$ method algebraically measures the "quality" of an opponent.
- **Networks and Graphs:** Dominance matrices are adjacency matrices for directed graphs. Squaring the matrix algebraically counts the number of 2-step paths between nodes in a network.
- **Real-World Algorithms:** This logic is the foundational mathematics behind modern search engines (like Google's PageRank algorithm), which rank web pages based on the quantity and quality of incoming links.

2.10 Leslie Matrices

Matrices can model dynamic, real-world systems that change over time. In biology and ecology, the **Leslie matrix** is a standard model used to predict the population growth of organisms by breaking the population down into distinct age groups (classes). Because females typically drive population reproduction rates, Leslie matrices generally only track the female portion of a population.

The Structure of a Leslie Matrix

Suppose a population is divided into n age classes of equal time duration.

- Let f_i be the **fecundity (fertility) rate**: the average number of female offspring born to a female in age class i .
- Let s_i be the **survival rate**: the probability that a female in age class i survives to enter age class $i + 1$.

The Leslie matrix L is a square $n \times n$ matrix built with a very specific layout:

- The **first row** contains the fecundity rates $(f_1, f_2, \dots, f_n)$.
- The **sub-diagonal** (the diagonal immediately below the main diagonal) contains the survival rates $(s_1, s_2, \dots, s_{n-1})$.
- All other entries are exactly zero.

For a 3-age-class population, the matrix looks like this:

$$L = \begin{bmatrix} f_1 & f_2 & f_3 \\ s_1 & 0 & 0 \\ 0 & s_2 & 0 \end{bmatrix}$$

Population Projection

If we represent the number of individuals in each age class at time t as a column matrix (a state vector) $\mathbf{n}_t$, we can find the population at the next time interval by multiplying by L :

$$\mathbf{n}_{t+1} = L\mathbf{n}_t$$

To project k time periods into the future from an initial population $\mathbf{n}_0$, we use matrix exponentiation:

$$\mathbf{n}_k = L^k \mathbf{n}_0$$

Problem 2.11: Projecting a Rodent Population

A species of rodent is divided into three age classes, each spanning one month: 0–1 month, 1–2 months, and 2–3 months. The maximum lifespan is 3 months.

- **Fertility rates:** Class 1 females produce 0 offspring. Class 2 females produce 2.0 offspring. Class 3 females produce 1.5 offspring.
- **Survival rates:** 40% (0.4) of Class 1 survive to Class 2. 30% (0.3) of Class 2 survive to Class 3.
- **Initial Population ($\mathbf{n}_0$):** There are currently 100 females in Class 1, 50 in Class 2, and 20 in Class 3.

- (a) Construct the Leslie matrix, L , for this population.
- (b) Calculate the population state vector after 1 month ($\mathbf{n}_1$).
- (c) Calculate the population state vector after 2 months ($\mathbf{n}_2$).

Solution 2.11**(a) Construct the Leslie matrix**

Row 1 contains the fertility rates: 0, 2.0, 1.5.

The sub-diagonal contains the survival rates: 0.4 and 0.3.

$$L = \begin{bmatrix} 0 & 2.0 & 1.5 \\ 0.4 & 0 & 0 \\ 0 & 0.3 & 0 \end{bmatrix}$$

(b) Population after 1 month ($\mathbf{n}_1$)

Multiply the Leslie matrix by the initial population vector $\mathbf{n}_0$:

$$\begin{aligned} \mathbf{n}_1 = L\mathbf{n}_0 &= \begin{bmatrix} 0 & 2.0 & 1.5 \\ 0.4 & 0 & 0 \\ 0 & 0.3 & 0 \end{bmatrix} \begin{bmatrix} 100 \\ 50 \\ 20 \end{bmatrix} = \begin{bmatrix} 0(100) + 2.0(50) + 1.5(20) \\ 0.4(100) + 0(50) + 0(20) \\ 0(100) + 0.3(50) + 0(20) \end{bmatrix} \\ &= \begin{bmatrix} 0 + 100 + 30 \\ 40 \\ 15 \end{bmatrix} = \begin{bmatrix} 130 \\ 40 \\ 15 \end{bmatrix} \end{aligned}$$

After 1 month, there are 130 newborns, 40 adults (1-2 mo), and 15 elders (2-3 mo).

(c) Population after 2 months ($\mathbf{n}_2$)

Multiply the Leslie matrix by the new population vector $\mathbf{n}_1$:

$$\begin{aligned} \mathbf{n}_2 = L\mathbf{n}_1 &= \begin{bmatrix} 0 & 2.0 & 1.5 \\ 0.4 & 0 & 0 \\ 0 & 0.3 & 0 \end{bmatrix} \begin{bmatrix} 130 \\ 40 \\ 15 \end{bmatrix} = \begin{bmatrix} 0(130) + 2.0(40) + 1.5(15) \\ 0.4(130) + 0(40) + 0(15) \\ 0(130) + 0.3(40) + 0(15) \end{bmatrix} \\ &= \begin{bmatrix} 0 + 80 + 22.5 \\ 52 \\ 12 \end{bmatrix} = \begin{bmatrix} 102.5 \\ 52 \\ 12 \end{bmatrix} \end{aligned}$$

(Note: In mathematical biology, it is standard practice to keep decimals during intermediate calculation steps to avoid compounding rounding errors, and only round to whole animals when stating the final interpretive answer. So we have roughly 103 newborns, 52 adults, and 12 elders).

Takeaways 2.11

- **Structured Flow:** The rigid structure of the Leslie matrix ensures mathematical logic: the top row calculates all new births and drops them into age class 1, while the sub-diagonal pushes the survivors from class i down into class $i + 1$.
- **Iteration:** A system that updates based on its previous state ($\mathbf{n}_{t+1} = L\mathbf{n}_t$) is called a dynamical system. Matrices are perfect for iterating large blocks of data simultaneously.
- **Long-term Behavior:** While outside the scope of this topic, the long-term survival or extinction of a population depends entirely on the largest *eigenvalue* of its Leslie matrix!

3 Part 1: Worked Problems

3.1 Basic

Problem 3.1: School Demographics Matrix

Based on the demographic data provided in the school census: At a certain school, there are 110 girls and 110 boys in Year 7. The numbers of girls and boys in the other year levels are 120 and 105 in Year 8, 140 and 130 in Year 9, 100 and 95 in Year 10, 90 and 85 in Year 11, and 60 and 65 in Year 12.

- Summarise this information in a 6×2 matrix, S , where the rows represent the year levels (7 to 12) and the columns represent the gender (girls, boys).
- Define a column matrix C such that the matrix product SC results in a 6×1 matrix representing the total number of students in each year level. Calculate SC .
- Define a row matrix R such that the matrix product RS results in a 1×2 matrix representing the total number of girls and boys in the school overall. Calculate RS .
- The school receives an extracurricular grant of \$50 per girl and \$45 per boy. Write down a column matrix G representing these amounts, and calculate the product SG to find the total grant amount allocated to each year level.

Hint:

- **Part (a):** Organise the data sequentially. Ensure your rows correspond to Years 7 through 12, and the first and second columns correspond to girls and boys, respectively.
- **Part (b):** To sum the elements across the rows of a matrix using matrix multiplication, you need to post-multiply by a column matrix of 1s. Check your dimensions: a (6×2) matrix multiplied by a (2×1) matrix yields a (6×1) matrix.
- **Part (c):** To sum the elements down the columns, you must pre-multiply by a row matrix of 1s. Verify the dimensions: (1×6) multiplied by (6×2) .
- **Part (d):** Matrix multiplication pairs the row elements of the first matrix with the column elements of the second matrix. Make sure the \$50 aligns with the girls' column and the \$45 aligns with the boys' column.

Solution 3.1

(a)

$$S = \begin{bmatrix} 110 & 110 \\ 120 & 105 \\ 140 & 130 \\ 100 & 95 \\ 90 & 85 \\ 60 & 65 \end{bmatrix}$$

(b)

$$C = \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \quad SC = \begin{bmatrix} 110 & 110 \\ 120 & 105 \\ 140 & 130 \\ 100 & 95 \\ 90 & 85 \\ 60 & 65 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 220 \\ 225 \\ 270 \\ 195 \\ 175 \\ 125 \end{bmatrix}$$

(c)

$$R = [1 \quad 1 \quad 1 \quad 1 \quad 1 \quad 1]$$

$$RS = [1 \quad 1 \quad 1 \quad 1 \quad 1 \quad 1] \begin{bmatrix} 110 & 110 \\ 120 & 105 \\ 140 & 130 \\ 100 & 95 \\ 90 & 85 \\ 60 & 65 \end{bmatrix} = [620 \quad 590]$$

(d)

$$G = \begin{bmatrix} 50 \\ 45 \end{bmatrix}, \quad SG = \begin{bmatrix} 110 & 110 \\ 120 & 105 \\ 140 & 130 \\ 100 & 95 \\ 90 & 85 \\ 60 & 65 \end{bmatrix} \begin{bmatrix} 50 \\ 45 \end{bmatrix} = \begin{bmatrix} 10450 \\ 10725 \\ 12850 \\ 9275 \\ 8325 \\ 5925 \end{bmatrix}$$

Takeaways 3.1

- **Data Organisation:** Matrices are powerful tools for neatly organising and storing real-world demographic data.
- **Summation Vectors:** Matrix multiplication using vectors of 1s is an efficient algebraic method for summing rows or columns of a data matrix without writing out lengthy addition equations.
- **Dimensionality Checks:** Always verify conformability before multiplying matrices. The number of columns in the leading matrix must equal the number of rows in the trailing matrix (e.g., $m \times n$ multiplied by $n \times p$ gives an $m \times p$ matrix).

3.2 Medium**Problem 3.2: Matrix Properties and Transformations**

Consider a general 2×2 matrix $A = \begin{bmatrix} p & q \\ r & s \end{bmatrix}$ and its associated adjugate matrix $A^* = \begin{bmatrix} s & -q \\ -r & p \end{bmatrix}$.

- Evaluate the matrix product AA^* and express the result in the form kI , where I is the 2×2 identity matrix and k is a scalar expressed in terms of p, q, r , and s .
- Given that $k \neq 0$, use your result from part (a) to algebraically deduce the standard expression for the inverse matrix, A^{-1} .
- Evaluate the reverse product A^*A and hence show that for this specific pair of matrices, multiplication is commutative.
- A linear transformation in the plane is represented by the matrix $M = \begin{bmatrix} 2x & 3 \\ 8 & x \end{bmatrix}$.
Using the concept of k from part (a), determine the exact possible values of x for which this transformation cannot be reversed (i.e., the matrix is singular).

Hint:

- **Part (a):** Perform standard row-by-column matrix multiplication. Expand the terms, simplify the diagonals, and factor out any common algebraic expressions to reveal the identity matrix.
- **Part (b):** Recall that the definition of an inverse matrix is $AA^{-1} = I$. Take your equation from part (a), $AA^* = kI$, and manipulate it algebraically to isolate I on one side of the equation.
- **Part (c):** Multiply the matrices again, but this time with A^* on the left and A on the right. Compare the resulting matrix to the one you found in part (a).
- **Part (d):** A matrix transformation does not have an inverse when its associated scalar k (the determinant) is equal to zero. Set up an equation for $k = 0$ using the elements of matrix M and solve for x .

Solution 3.2

(a)

$$AA^* = \begin{bmatrix} p & q \\ r & s \end{bmatrix} \begin{bmatrix} s & -q \\ -r & p \end{bmatrix} = \begin{bmatrix} ps - qr & -pq + qp \\ rs - sr & -rq + sp \end{bmatrix} = \begin{bmatrix} ps - qr & 0 \\ 0 & ps - qr \end{bmatrix}$$

$$\text{Factoring out the common term: } AA^* = (ps - qr) \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = (ps - qr)I.$$

Therefore, $k = ps - qr$.

(b)

From part (a), we have $AA^* = (ps - qr)I$.Assuming $ps - qr \neq 0$, we can divide both sides by this scalar:

$$A \left(\frac{1}{ps - qr} A^* \right) = I$$

$$\text{Since } AA^{-1} = I, \text{ it follows that } A^{-1} = \frac{1}{ps - qr} \begin{bmatrix} s & -q \\ -r & p \end{bmatrix}.$$

(c)

$$A^*A = \begin{bmatrix} s & -q \\ -r & p \end{bmatrix} \begin{bmatrix} p & q \\ r & s \end{bmatrix} = \begin{bmatrix} sp - qr & sq - qs \\ -rp + pr & -rq + ps \end{bmatrix} = \begin{bmatrix} ps - qr & 0 \\ 0 & ps - qr \end{bmatrix} = (ps - qr)I$$

Since $AA^* = (ps - qr)I$ and $A^*A = (ps - qr)I$, we have shown that $AA^* = A^*A$. Thus, multiplication of a matrix by its adjugate is commutative.

(d)

For the transformation to have no inverse, $k = ps - qr = 0$.For matrix M , $p = 2x$, $q = 3$, $r = 8$, and $s = x$.

$$k = (2x)(x) - (3)(8) = 2x^2 - 24$$

$$\text{Setting } k = 0: 2x^2 - 24 = 0 \implies x^2 = 12 \implies x = \pm\sqrt{12} = \pm 2\sqrt{3}.$$

Takeaways 3.2

- **Determinant:** The scalar value $k = ps - qr$ derived in part (a) is the determinant of a 2×2 matrix. It is the fundamental factor that dictates whether a matrix has an inverse.
- **Commutativity of Inverses:** While general matrix multiplication is strictly non-commutative ($AB \neq BA$), a matrix and its adjugate (or inverse) are a special exception where multiplication order does not change the result.
- **Singular Matrices:** Matrices with a determinant of zero are called singular. Geometrically, these matrices represent transformations that collapse 2D space into a 1D line or a point, which is why the mapping cannot be algebraically reversed.

Problem 3.3: Matrix Algebra and Commutativity

Consider general 2×2 matrices A and B .

- (a) By expanding the product, show that $(A + B)^2 = A^2 + AB + BA + B^2$.
- (b) Hence, state the necessary mathematical condition for matrices A and B such that the standard binomial expansion $(A + B)^2 = A^2 + 2AB + B^2$ is valid.
- (c) Let $A = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$ representing a reflection in the x-axis, and $B = \begin{bmatrix} k & 0 \\ 0 & k \end{bmatrix}$ representing a dilation by scale factor k . Verify algebraically that for these specific matrices, $(A + B)^2 = A^2 + 2AB + B^2$.
- (d) Let $C = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ and $D = \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix}$. Evaluate $(C + D)^2$ and $C^2 + 2CD + D^2$, demonstrating they are not equal, and explain why this algebraic difference occurs in terms of matrix multiplication properties.

Hint:

- **Part (a):** Write $(A + B)^2$ as $(A + B)(A + B)$ and apply the distributive law carefully, ensuring you maintain the exact left-to-right order of matrix multiplication.
- **Part (b):** Compare your expanded result from part (a) to the target expression $A^2 + 2AB + B^2$. What must the middle terms $AB + BA$ simplify to?
- **Part (c):** Calculate the individual products AB and BA first. Check if they satisfy the condition you identified in part (b).
- **Part (d):** For the first expression, perform the matrix addition $C + D$ before squaring the result. For the second expression, calculate C^2 , D^2 , and CD separately before summing them.

Solution 3.3

(a)

$$(A+B)^2 = (A+B)(A+B) = A(A+B) + B(A+B) = A^2 + AB + BA + B^2.$$

(b)

For $A^2 + AB + BA + B^2$ to equal $A^2 + 2AB + B^2$, we must have:

$$AB + BA = 2AB \implies BA = AB.$$

The necessary condition is that matrices A and B must commute.

(c)

$$AB = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} k & 0 \\ 0 & k \end{bmatrix} = \begin{bmatrix} k & 0 \\ 0 & -k \end{bmatrix}$$

$$BA = \begin{bmatrix} k & 0 \\ 0 & k \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} = \begin{bmatrix} k & 0 \\ 0 & -k \end{bmatrix}$$

Since $AB = BA$, the matrices commute, and therefore the binomial expansion $(A+B)^2 = A^2 + 2AB + B^2$ holds valid for this specific pair.

(d)

$$C + D = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} + \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 2 & 0 \end{bmatrix}.$$

$$(C + D)^2 = \begin{bmatrix} 0 & 0 \\ 2 & 0 \end{bmatrix} \begin{bmatrix} 0 & 0 \\ 2 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}.$$

Next, evaluating the components of the expanded form:

$$C^2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I \text{ and } D^2 = \begin{bmatrix} -1 & 0 \\ 0 & -1 \end{bmatrix} = -I.$$

$$CD = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 0 & -1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}.$$

$$C^2 + 2CD + D^2 = I + 2 \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} - I = \begin{bmatrix} 2 & 0 \\ 0 & -2 \end{bmatrix}.$$

Since $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix} \neq \begin{bmatrix} 2 & 0 \\ 0 & -2 \end{bmatrix}$, the expressions are unequal. This occurs because general matrix multiplication is non-commutative ($CD \neq DC$), preventing the cross-terms from simplifying to $2CD$.

Takeaways 3.3

- **Matrix Non-Commutativity:** The most critical difference between standard real number algebra and matrix algebra is that matrix multiplication is generally non-commutative ($AB \neq BA$).
- **Binomial Expansion:** Standard polynomial identities like $(x+y)^2 = x^2 + 2xy + y^2$ cannot be blindly applied to matrices unless the specific matrices are proven to commute.
- **Geometric Connection:** The algebraic failure of commutativity mirrors the geometric reality of transformations. For instance, applying a reflection then a rotation often results in a different final orientation compared to applying the rotation then the reflection.

3.3 Advanced

Problem 3.4: Matrix Inverses and Matrix Equations

Let A and B be 2×2 matrices defined as $A = \begin{bmatrix} k & 2 \\ -1 & 1 \end{bmatrix}$ and $B = \begin{bmatrix} 1 & -1 \\ 2 & 3 \end{bmatrix}$, where k is a real constant.

- Determine the determinant of A in terms of k , and hence state the value of k for which matrix A is singular.
- Let $k = 3$. Determine the inverse matrices A^{-1} and B^{-1} .
- For $k = 3$, calculate the matrix product AB and hence determine $(AB)^{-1}$.
- Evaluate the product $B^{-1}A^{-1}$ to verify the property $(AB)^{-1} = B^{-1}A^{-1}$. Hence, use this property to make the matrix X the subject of the matrix equation $ABX = C$, where C is a 2×2 matrix.

Hint:

- **Part (a):** The determinant of a 2×2 matrix $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ is $ad - bc$. A matrix is singular (has no inverse) when its determinant equals zero.

- **Part (b):** Use the formula for the inverse of a 2×2 matrix: $\frac{1}{\det(M)} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$.

- **Part (c):** Multiply matrix A by matrix B using the standard row-by-column method. Then find the inverse of the resulting 2×2 matrix.

- **Part (d):** Multiply B^{-1} (on the left) by A^{-1} (on the right) and factor out the scalar multipliers. To solve $ABX = C$, remember that you must multiply both sides of the equation by the inverse of AB on the left.

Solution 3.4**(a)**

$$\det(A) = (k)(1) - (2)(-1) = k + 2.$$

For matrix A to be singular, $\det(A) = 0$. Therefore, $k = -2$.

(b)

Given $k = 3$, $A = \begin{bmatrix} 3 & 2 \\ -1 & 1 \end{bmatrix}$.

$$\det(A) = 3 + 2 = 5. \text{ Thus, } A^{-1} = \frac{1}{5} \begin{bmatrix} 1 & -2 \\ 1 & 3 \end{bmatrix}.$$

$$\det(B) = (1)(3) - (-1)(2) = 3 + 2 = 5. \text{ Thus, } B^{-1} = \frac{1}{5} \begin{bmatrix} 3 & 1 \\ -2 & 1 \end{bmatrix}.$$

(c)

$$AB = \begin{bmatrix} 3 & 2 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} 1 & -1 \\ 2 & 3 \end{bmatrix} = \begin{bmatrix} 3(1) + 2(2) & 3(-1) + 2(3) \\ -1(1) + 1(2) & -1(-1) + 1(3) \end{bmatrix} = \begin{bmatrix} 7 & 3 \\ 1 & 4 \end{bmatrix}.$$

$$\det(AB) = (7)(4) - (3)(1) = 28 - 3 = 25.$$

$$(AB)^{-1} = \frac{1}{25} \begin{bmatrix} 4 & -3 \\ -1 & 7 \end{bmatrix}.$$

(d)

$$B^{-1}A^{-1} = \left(\frac{1}{5} \begin{bmatrix} 3 & 1 \\ -2 & 1 \end{bmatrix} \right) \left(\frac{1}{5} \begin{bmatrix} 1 & -2 \\ 1 & 3 \end{bmatrix} \right) = \frac{1}{25} \begin{bmatrix} 3(1) + 1(1) & 3(-2) + 1(3) \\ -2(1) + 1(1) & -2(-2) + 1(3) \end{bmatrix} = \frac{1}{25} \begin{bmatrix} 4 & -3 \\ -1 & 7 \end{bmatrix}.$$

This verifies that $(AB)^{-1} = B^{-1}A^{-1}$.

To solve $ABX = C$:

$$(AB)^{-1}(AB)X = (AB)^{-1}C$$

$$IX = (AB)^{-1}C$$

$$X = B^{-1}A^{-1}C.$$

Takeaways 3.4

- **The "Socks and Shoes" Property:** The inverse of a matrix product reverses the order of the original matrices: $(AB)^{-1} = B^{-1}A^{-1}$.
- **Matrix Algebra in Equations:** When isolating a matrix variable (like X in $ABX = C$), you must respect the non-commutative nature of matrices by multiplying by the inverse on the correct side (in this case, pre-multiplying on the left).
- **Determinant of a Product:** Notice that $\det(AB) = 25$, while $\det(A) = 5$ and $\det(B) = 5$. This verifies another key property: $\det(AB) = \det(A) \times \det(B)$.

Problem 3.5: Matrix Inverses and the Zero Matrix

Let A, B, C , and D be 2×2 matrices, where O is the zero matrix and I is the identity matrix.

- (a) Assume A is an invertible matrix and that $AB = O$. Prove algebraically that B must be the zero matrix.
- (b) Suppose instead that C and D are both strictly non-zero matrices, yet their product is the zero matrix ($CD = O$). By referencing your proof in part (a), deduce whether C can be an invertible matrix. Support your answer with logical reasoning.
- (c) A matrix M is defined as being self-inverse if $M^{-1} = M$. By showing this implies $M^2 = I$, determine the algebraic condition on real numbers a, b , and c such that $M = \begin{bmatrix} a & b \\ c & -a \end{bmatrix}$ is a self-inverse matrix.
- (d) Using your condition from part (c), construct a specific matrix M where $a = 3$, and b and c are integers. Verify that your chosen matrix is self-inverse by explicitly evaluating M^2 .

Hint:

- **Part (a):** Start with the given equation $AB = O$. Since A is invertible, you are permitted to pre-multiply both sides of the equation by A^{-1} . What does $A^{-1}A$ simplify to?
- **Part (b):** Use a proof by contradiction. What would happen if C was invertible? Apply the exact same logic established in part (a) and compare the result to the condition that D is a non-zero matrix.
- **Part (c):** Multiply both sides of $M^{-1} = M$ by M . Then, square the given matrix $M = \begin{bmatrix} a & b \\ c & -a \end{bmatrix}$ using standard row-by-column multiplication and equate the resulting elements to the corresponding elements of the identity matrix, I .
- **Part (d):** Substitute $a = 3$ into your condition from part (c). Find two integers b and c that multiply together to satisfy the equation, construct the matrix, and manually multiply $M \times M$ to check it yields I .

Solution 3.5**(a)**

Given $AB = O$ and A is invertible, A^{-1} exists. Pre-multiplying both sides by A^{-1} :

$$A^{-1}(AB) = A^{-1}O$$

$$(A^{-1}A)B = O$$

$$IB = O \implies B = O.$$

(b)

Assume for the sake of contradiction that C is invertible.

Then, given $CD = O$, we could pre-multiply by C^{-1} to get $C^{-1}(CD) = C^{-1}O \implies D = O$.

However, the problem states that D is a non-zero matrix. This is a contradiction. Therefore, the assumption that C is invertible must be false; C must be a singular matrix.

(c)

If $M^{-1} = M$, then multiplying both sides by M gives $MM^{-1} = MM \implies I = M^2$.

Evaluating M^2 :

$$M^2 = \begin{bmatrix} a & b \\ c & -a \end{bmatrix} \begin{bmatrix} a & b \\ c & -a \end{bmatrix} = \begin{bmatrix} a^2 + bc & ab - ab \\ ac - ac & bc + (-a)^2 \end{bmatrix} = \begin{bmatrix} a^2 + bc & 0 \\ 0 & a^2 + bc \end{bmatrix}.$$

For $M^2 = I = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$, we must have the condition: $a^2 + bc = 1$.

(d)

Let $a = 3$. From part (c), $3^2 + bc = 1 \implies 9 + bc = 1 \implies bc = -8$.

Choose integers $b = 2$ and $c = -4$ (other integer pairs like $4, -2$ are also valid).

$$M = \begin{bmatrix} 3 & 2 \\ -4 & -3 \end{bmatrix}.$$

Verification:

$$M^2 = \begin{bmatrix} 3 & 2 \\ -4 & -3 \end{bmatrix} \begin{bmatrix} 3 & 2 \\ -4 & -3 \end{bmatrix} = \begin{bmatrix} 9 - 8 & 6 - 6 \\ -12 + 12 & -8 + 9 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = I.$$

Takeaways 3.5

- **Zero Product Property Failure:** In real number algebra, $xy = 0$ implies $x = 0$ or $y = 0$. In matrix algebra, $CD = O$ does *not* necessarily mean $C = O$ or $D = O$. Non-zero matrices can multiply to give the zero matrix.
- **Singularity is Key:** For two non-zero matrices to multiply to the zero matrix, they must both be singular (determinant of zero). If either were invertible, the other would be forced to be the zero matrix.
- **Self-Inverse Matrices:** Matrices where $M^{-1} = M$ exist in infinitely many non-trivial forms beyond just the identity matrix. Geometrically, self-inverse (or involutory) matrices often represent reflections, where applying the transformation twice returns all coordinates to their original positions.

Problem 3.6: Matrix Methods for Simultaneous Equations

Consider the following system of simultaneous linear equations with real parameter k :

$$(k - 1)x + 3y = k + 2$$

$$2x + (k - 2)y = 4$$

- (a) Express this system as a matrix equation of the form $AX = B$, and find a simplified quadratic expression for the determinant of the coefficient matrix A .
- (b) Determine the values of k for which the matrix A is singular (i.e., does not have an inverse).
- (c) For each value of k found in part (b), determine whether the system has no solutions or infinitely many solutions. Provide a brief geometric justification for your answers.
- (d) Assuming the matrix A is invertible, use the matrix inverse method to find the unique solution for x and y in terms of k , expressing your final answers in their simplest rational form.

Hint:

- **Part (a):** The coefficient matrix A contains the multipliers for x and y . The determinant of a 2×2 matrix $\begin{bmatrix} a & b \\ c & d \end{bmatrix}$ is calculated as $ad - bc$.

- **Part (b):** A matrix is singular when its determinant is exactly zero. Set your expression from part (a) to zero and solve the resulting quadratic equation for k .

- **Part (c):** Substitute your k values back into the original system of equations. If the two equations represent distinct parallel lines (e.g., yielding a false statement like $0 = 5$), there is no solution. If they represent the exact same line, there are infinitely many solutions.

- **Part (d):** Use the inverse matrix formula $X = A^{-1}B = \frac{1}{\det(A)} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix} B$. After multiplying the matrices, factor the resulting numerators to see if they cancel with the determinant in the denominator.

Solution 3.6

(a)

$$\text{Matrix equation: } \begin{bmatrix} k-1 & 3 \\ 2 & k-2 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} k+2 \\ 4 \end{bmatrix}$$

$$\det(A) = (k-1)(k-2) - (3)(2) = k^2 - 3k + 2 - 6 = k^2 - 3k - 4.$$

$$\text{Factored form: } \det(A) = (k-4)(k+1).$$

(b)

Matrix A is singular when $\det(A) = 0$.

$$k^2 - 3k - 4 = 0 \implies (k-4)(k+1) = 0.$$

Therefore, A is singular when $k = 4$ or $k = -1$.

(c)

Case $k = 4$:Equations become $3x + 3y = 6$ (which simplifies to $x + y = 2$) and $2x + 2y = 4$ (which simplifies to $x + y = 2$).Because the equations represent coincident (identical) lines, there are **infinitely many solutions**.**Case $k = -1$:**Equations become $-2x + 3y = 1$ and $2x - 3y = 4$.Adding the two equations yields $0 = 5$, which is mathematically inconsistent. Because the equations represent distinct parallel lines, there are **no solutions**.

(d)

For $k \neq 4$ and $k \neq -1$, A is invertible.

$$A^{-1} = \frac{1}{(k-4)(k+1)} \begin{bmatrix} k-2 & -3 \\ -2 & k-1 \end{bmatrix}$$

$$X = \frac{1}{(k-4)(k+1)} \begin{bmatrix} k-2 & -3 \\ -2 & k-1 \end{bmatrix} \begin{bmatrix} k+2 \\ 4 \end{bmatrix}$$

$$x = \frac{(k-2)(k+2)-12}{(k-4)(k+1)} = \frac{k^2-4-12}{(k-4)(k+1)} = \frac{k^2-16}{(k-4)(k+1)} = \frac{(k-4)(k+4)}{(k-4)(k+1)} = \frac{k+4}{k+1}$$

$$y = \frac{-2(k+2)+4(k-1)}{(k-4)(k+1)} = \frac{-2k-4+4k-4}{(k-4)(k+1)} = \frac{2k-8}{(k-4)(k+1)} = \frac{2(k-4)}{(k-4)(k+1)} = \frac{2}{k+1}$$

$$\text{Unique solution: } x = \frac{k+4}{k+1} \text{ and } y = \frac{2}{k+1}.$$

Takeaways 3.6

- **Determinant as a Geometric Indicator:** Just as the discriminant reveals the nature of roots in a quadratic, the determinant of a coefficient matrix dictates the geometry of linear systems. A non-zero determinant guarantees intersecting lines (one unique solution).
- **Interpreting Singularity:** A singular matrix ($\det = 0$) means the matrix transformation collapses 2D space. For a system of equations, this means the lines share the exact same gradient, resulting in either parallel lines (no intersection) or coincident lines (infinite intersections).
- **Algebraic Simplification:** When solving parametrically with the matrix inverse method, finding the solution often involves polynomial division or factoring. The non-zero roots of the determinant will frequently appear as cancellable factors in the numerators of your x and y expressions.

Problem 3.7: Determinant Properties and Matrix Algebra

Let $A = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ and $B = \begin{bmatrix} p & q \\ r & s \end{bmatrix}$ be arbitrary 2×2 matrices.

- (a) Prove algebraically that $\det(AB) = \det(A)\det(B)$.
- (b) Using the result from part (a), prove that if a matrix M is invertible, then $\det(M^{-1}) = \frac{1}{\det(M)}$.
- (c) Consider the matrices $C = \begin{bmatrix} k & 2 \\ 1 & -1 \end{bmatrix}$ and $D = \begin{bmatrix} 2 & 0 \\ -1 & k \end{bmatrix}$, where k is a non-zero real constant. Using the determinant properties established above, determine the exact values of k for which $\det(C^2 D^{-1}) = \frac{9}{2}$.
- (d) Using the larger value of k found in part (c), calculate $\det(C + D)$ and explicitly show that $\det(C + D) \neq \det(C) + \det(D)$.

Hint:

- **Part (a):** Multiply the matrices A and B first using standard row-by-column multiplication. Then apply the $ad - bc$ determinant rule to the resulting matrix. Expand all terms and factor them to show the expression matches $(ad - bc)(ps - qr)$.
- **Part (b):** Start with the fundamental definition of an inverse matrix: $MM^{-1} = I$. Apply the determinant operator to both sides of this equation.
- **Part (c):** Apply the rules $\det(C^2) = (\det(C))^2$ and $\det(D^{-1}) = 1/\det(D)$. Form an algebraic equation purely in terms of k and solve the resulting quadratic.
- **Part (d):** Add the matrices C and D element-by-element first to find the matrix $(C + D)$, then find its determinant. Compare this final value to the sum of their individual determinants.

Solution 3.7

(a)

$$AB = \begin{bmatrix} a & b \\ c & d \end{bmatrix} \begin{bmatrix} p & q \\ r & s \end{bmatrix} = \begin{bmatrix} ap + br & aq + bs \\ cp + dr & cq + ds \end{bmatrix}.$$

$$\det(AB) = (ap + br)(cq + ds) - (aq + bs)(cp + dr)$$

$$\det(AB) = (apcq + apds + brcq + brds) - (aqcp + aqdr + bscp + bsdr)$$

The terms $apcq$ and $bsdr$ cancel out:

$$\det(AB) = apds + brcq - aqdr - bscp$$

Factoring by grouping:

$$\det(AB) = ad(ps - qr) - bc(ps - qr) = (ad - bc)(ps - qr)$$

Since $\det(A) = ad - bc$ and $\det(B) = ps - qr$, we have proven $\det(AB) = \det(A)\det(B)$.

(b)

By definition, if M is invertible, $MM^{-1} = I$.

Taking the determinant of both sides: $\det(MM^{-1}) = \det(I)$.

Using the property from part (a): $\det(M)\det(M^{-1}) = 1$.

Because M is invertible, $\det(M) \neq 0$. Thus, we can divide by $\det(M)$ to yield $\det(M^{-1}) = \frac{1}{\det(M)}$.

(c)

$$\det(C) = (k)(-1) - (2)(1) = -k - 2.$$

$$\det(D) = (2)(k) - (0)(-1) = 2k.$$

$$\text{Using the properties: } \det(C^2D^{-1}) = \det(C)^2 \times \det(D^{-1}) = \frac{\det(C)^2}{\det(D)}.$$

$$\frac{(-k-2)^2}{2k} = \frac{9}{2} \implies \frac{k^2+4k+4}{2k} = \frac{9}{2}$$

$$\text{Cross-multiplying: } 2(k^2 + 4k + 4) = 18k \implies 2k^2 + 8k + 8 = 18k$$

$$2k^2 - 10k + 8 = 0 \implies k^2 - 5k + 4 = 0.$$

Factoring gives $(k - 4)(k - 1) = 0$, so $k = 4$ or $k = 1$.

(d)

Using the larger value $k = 4$:

$$C = \begin{bmatrix} 4 & 2 \\ 1 & -1 \end{bmatrix} \implies \det(C) = -4 - 2 = -6.$$

$$D = \begin{bmatrix} 2 & 0 \\ -1 & 4 \end{bmatrix} \implies \det(D) = 8 - 0 = 8.$$

$$\det(C) + \det(D) = -6 + 8 = 2.$$

Now, finding $C + D$:

$$C + D = \begin{bmatrix} 4+2 & 2+0 \\ 1-1 & -1+4 \end{bmatrix} = \begin{bmatrix} 6 & 2 \\ 0 & 3 \end{bmatrix}.$$

$$\det(C + D) = (6)(3) - (2)(0) = 18.$$

Since $18 \neq 2$, we have shown that $\det(C + D) \neq \det(C) + \det(D)$.

Takeaways 3.7

- **Multiplicative Nature of Determinants:** The determinant is a strictly multiplicative function. Matrix exponents and inverses can be evaluated directly through their determinants without calculating the resulting complex matrices first.
- **Non-Linearity of Determinants:** The determinant is *not* a linear operator. $\det(A + B)$ is rarely equal to $\det(A) + \det(B)$.
- **Algebraic Proofs:** Performing brute-force algebraic expansions on 2×2 matrices (as in part a) is a foundational QCAA syllabus skill for proving matrix properties from first principles.

Problem 3.8: Dominance Arrays and Tournament Rankings

Five schools—A, B, C, D, and E—compete in a round-robin hockey tournament where no matches end in a draw. The results are as follows:

- School A defeated B, C, and D.
 - School B defeated C and D.
 - School C defeated D and E.
 - School D defeated E.
 - School E defeated A and B.
- (a) Construct the 5×5 dominance array, M , for this tournament, where a 1 in row i and column j indicates that team i defeated team j .
- (b) Evaluate the squared array M^2 . What does the numerical value located in row 5, column 3 of M^2 represent contextually?
- (c) The tournament committee decides to rank the schools using a total power score vector P , calculated by the formula $P = (M + M^2)\mathbf{v}$, where $\mathbf{v}$ is a 5×1 column vector of ones. Calculate P and state the final ranking of the five schools.
- (d) For any standard $n \times n$ round-robin dominance array (where there are no draws), the sum of all elements in the array is a fixed constant depending only on n . Determine this algebraic expression in terms of n , and provide a brief logical justification based on the properties of a round-robin tournament.

Hint:

- **Part (a):** Set up a 5×5 grid. If A beats B, place a 1 in the row for A and column for B, and a 0 in the row for B and column for A. The main diagonal will contain only zeros.
- **Part (b):** Multiply array M by itself using standard row-by-column multiplication. The value in row 5, column 3 corresponds to the pathways from E to C.
- **Part (c):** Adding M and M^2 combines the 1-step and 2-step dominance scores. Multiplying by a column vector of ones simply sums the rows of the resulting array.
- **Part (d):** In a round-robin tournament, how many total games are played between n teams? Remember that each match results in exactly one '1' being placed somewhere in the dominance array.

Solution 3.8

(a)

$$M = \begin{bmatrix} 0 & 1 & 1 & 1 & 0 \\ 0 & 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 & 1 \\ 1 & 1 & 0 & 0 & 0 \end{bmatrix}$$

(b)

$$M^2 = \begin{bmatrix} 0 & 0 & 1 & 2 & 2 \\ 0 & 0 & 0 & 1 & 2 \\ 1 & 1 & 0 & 0 & 1 \\ 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 2 & 2 & 0 \end{bmatrix}$$

The value in row 5, column 3 is 2. This represents the number of 2-step dominances School E has over School C (i.e., School E defeated two teams—A and B—who both subsequently defeated School C).

(c)

$$M + M^2 = \begin{bmatrix} 0 & 1 & 2 & 3 & 2 \\ 0 & 0 & 1 & 2 & 2 \\ 1 & 1 & 0 & 1 & 2 \\ 1 & 1 & 0 & 0 & 1 \\ 1 & 2 & 2 & 2 & 0 \end{bmatrix}$$

Summing the rows yields the total power score vector P :

$$P = \begin{bmatrix} 8 \\ 5 \\ 5 \\ 3 \\ 7 \end{bmatrix}$$

The rankings are: 1st: A, 2nd: E, Tied 3rd: B and C, 5th: D.

(d)

In an n -team round-robin tournament with no draws, every pair of teams plays exactly one match. The number of unique pairs is given by the combination formula $\frac{n(n-1)}{2}$. Since each match contributes exactly one '1' to the dominance array (and '0' to the reverse), the total sum of all elements in the array is exactly $\frac{n(n-1)}{2}$.

Takeaways 3.8

- **2-Step Dominance:** Squaring a dominance array reveals indirect victories, which are crucial for breaking ties or uncovering hidden strengths in a round-robin format.
- **Power Scores:** The expression $(M + M^2)\mathbf{v}$ is a standard algebraic method for calculating a comprehensive ranking score, weighing direct and indirect wins equally.
- **Array Properties:** Recognizing general algebraic properties of tournament structures (like the total sum of elements being $\frac{n(n-1)}{2}$) connects discrete combinatorics with array operations.

Problem 3.9: Dominance Matrices and Tournament Ratings

Five members of the Onyx Positional chess club—A, B, C, D, and E—compete in a round-robin tournament where no games end in a draw. The results of the matches are as follows:

- Player A defeated B and D.
 - Player B defeated C and D.
 - Player C defeated A and E.
 - Player D defeated E.
 - Player E defeated A and B.
- (a) Construct the 5×5 one-step dominance matrix, D , for this tournament, where a 1 in row i and column j indicates that player i defeated player j .
- (b) Calculate the two-step dominance matrix, D^2 . What does the numerical value located in row 3, column 2 of D^2 represent in the context of this chess tournament?
- (c) The club director decides to rank the players using a weighted total power score vector P , calculated by the formula $P = (D + 0.5D^2)\mathbf{v}$, where $\mathbf{v}$ is a 5×1 column vector of ones. Calculate P and state the final ranking of the five players.
- (d) A new player, F, joins the club late, plays against all five original members, and loses every game. Without recalculating the new 6×6 matrices from scratch, logically deduce how the addition of player F affects the relative total power scores (P) and the final rankings of players A through E from part (c).

Hint:

- **Part (a):** Set up a 5×5 grid. If A beats B, place a 1 in row A, column B, and a 0 in row B, column A. The main diagonal will contain only zeros.
- **Part (b):** Multiply matrix D by itself using standard row-by-column multiplication. The resulting values indicate the number of indirect pathways (2-step dominances) between players.
- **Part (c):** Adding D and $0.5D^2$ combines the direct wins with half-weighted indirect wins. Multiplying by a column vector of ones simply computes the sum of the rows of the resulting weighted matrix.
- **Part (d):** Consider how F's performance alters the row sums. F contributes a direct win (1-step dominance) to every other player. Does F contribute to anyone's 2-step dominance score, given that F defeated no one?

Solution 3.9

(a)

$$D = \begin{bmatrix} 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 \\ 1 & 1 & 0 & 0 & 0 \end{bmatrix}$$

(b)

$$D^2 = \begin{bmatrix} 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 \\ 1 & 1 & 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 \\ 1 & 1 & 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 & 1 & 1 \\ 1 & 0 & 0 & 1 & 2 \\ 1 & 2 & 0 & 1 & 0 \\ 1 & 1 & 0 & 0 & 0 \\ 0 & 1 & 1 & 2 & 0 \end{bmatrix}$$

The value in row 3, column 2 is 2. This represents the number of 2-step dominances Player C has over Player B. Specifically, Player C defeated two players (A and E), who both subsequently defeated Player B.

(c)

$$D + 0.5D^2 = \begin{bmatrix} 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 0 & 1 \\ 0 & 0 & 0 & 0 & 1 \\ 1 & 1 & 0 & 0 & 0 \end{bmatrix} + \begin{bmatrix} 0 & 0 & 0.5 & 0.5 & 0.5 \\ 0.5 & 0 & 0 & 0.5 & 1 \\ 0.5 & 1 & 0 & 0.5 & 0 \\ 0.5 & 0.5 & 0 & 0 & 0 \\ 0 & 0.5 & 0.5 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0.5 & 1.5 & 0.5 \\ 0.5 & 0 & 1 & 1.5 & 1 \\ 1.5 & 1 & 0 & 0.5 & 1 \\ 0.5 & 0.5 & 0 & 0 & 1 \\ 1 & 1.5 & 0.5 & 1 & 0 \end{bmatrix}$$

Summing the rows yields the total power score vector P :

$$P = \begin{bmatrix} 3.5 \\ 4.0 \\ 4.0 \\ 2.0 \\ 4.0 \end{bmatrix}$$

The rankings are: Tied 1st: Players B, C, and E (4.0). 4th: Player A (3.5). 5th: Player D (2.0).

(d)

Player F loses to everyone, so every player from A through E gains exactly 1 direct win (1-step dominance). Because Player F defeated no one, defeating F does not grant any indirect wins (2-step dominance) over anyone else. Thus, the 2-step dominance scores for players A through E remain unchanged.

In the formula $P = D + 0.5D^2$, every original player's score increases by exactly $1.0 + 0.5(0) = 1.0$.

Since every player's score increases by the exact same constant amount, the relative power scores and the final rankings of players A through E remain entirely unchanged.

Takeaways 3.9

- **Weighting Indirect Wins:** Standard dominance rankings often use $D + D^2$, but applying a fractional weight (like 0.5) to D^2 is a more realistic rating method that prioritizes direct head-to-head victories over indirect pathways.
- **Interpreting Matrix Elements:** The element (i, j) in D^2 physically counts the number of distinct intermediate opponents that player i defeated, who then went on to defeat player j .
- **Structural Additions to Graphs:** In discrete math, identifying how adding a "sink" node (a node with only incoming directed edges) affects the adjacency and reachability matrices allows you to bypass tedious brute-force calculations.

4 Part 2: Practice and Challenge Bank

4.1 Warm-Up Drills

4.2 Stretch Problems

Problem 4.1: Matrix Systems and Polynomial Curve Fitting

A quadratic function has a rule of the form $f(x) = ax^2 + bx + c$. It is known that the points $(-1, 12)$ and $(1, 4)$ are on the graph of f , alongside a third point $(2, 9)$.

- (a) By substituting the known points, write down three linear equations satisfied by a, b , and c . Formulate this information as a matrix equation of the form $M\mathbf{v} = \mathbf{k}$,

where $\mathbf{v} = \begin{bmatrix} a \\ b \\ c \end{bmatrix}$.

- (b) The inverse of the coefficient matrix M is given by $M^{-1} = \frac{1}{6} \begin{bmatrix} 1 & -3 & 2 \\ -3 & 3 & 0 \\ 2 & 6 & -2 \end{bmatrix}$. Use this inverse to solve the matrix equation and determine the exact values of a, b , and c .

- (c) Suppose the y -coordinate of the third point is changed to a parameter p , such that the point is now $(2, p)$. Write down the new constant vector $\mathbf{k}$, and use M^{-1} to determine the coefficients a, b , and c in terms of p .

- (d) Determine the specific value of p for which the quadratic function degenerates into a linear function (i.e., $a = 0$). Geometrically, what does this imply about the arrangement of the three points on the Cartesian plane?

Hint:

- **Part (a):** Substitute the x and y values of each coordinate pair into the quadratic equation to generate three distinct linear equations. The coefficients of a, b , and c will form your 3×3 matrix M .
- **Part (b):** To solve $M\mathbf{v} = \mathbf{k}$, pre-multiply both sides by the inverse matrix so that $\mathbf{v} = M^{-1}\mathbf{k}$.
- **Part (c):** Update the third entry of your $\mathbf{k}$ vector to p . Multiply the matrix M^{-1} by this new vector using standard row-by-column multiplication.
- **Part (d):** Take your parametric expression for a from part (c) and set it equal to 0. If a quadratic curve becomes a straight line, consider what that means for the points that define it.

Solution 4.1

(a)

Substituting $(-1, 12)$: $a(-1)^2 + b(-1) + c = 12 \implies a - b + c = 12$ Substituting $(1, 4)$: $a(1)^2 + b(1) + c = 4 \implies a + b + c = 4$ Substituting $(2, 9)$: $a(2)^2 + b(2) + c = 9 \implies 4a + 2b + c = 9$ Matrix equation:
$$\begin{bmatrix} 1 & -1 & 1 \\ 1 & 1 & 1 \\ 4 & 2 & 1 \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} 12 \\ 4 \\ 9 \end{bmatrix}.$$

(b)

$$\mathbf{v} = M^{-1}\mathbf{k} = \frac{1}{6} \begin{bmatrix} 1 & -3 & 2 \\ -3 & 3 & 0 \\ 2 & 6 & -2 \end{bmatrix} \begin{bmatrix} 12 \\ 4 \\ 9 \end{bmatrix}$$

$$a = \frac{1}{6}(1(12) - 3(4) + 2(9)) = \frac{1}{6}(12 - 12 + 18) = 3$$

$$b = \frac{1}{6}(-3(12) + 3(4) + 0(9)) = \frac{1}{6}(-36 + 12) = -4$$

$$c = \frac{1}{6}(2(12) + 6(4) - 2(9)) = \frac{1}{6}(24 + 24 - 18) = 5$$

Therefore, $a = 3$, $b = -4$, and $c = 5$.

(c)

With the new point $(2, p)$, the constant vector becomes $\mathbf{k} = \begin{bmatrix} 12 \\ 4 \\ p \end{bmatrix}$.

$$\mathbf{v} = \frac{1}{6} \begin{bmatrix} 1 & -3 & 2 \\ -3 & 3 & 0 \\ 2 & 6 & -2 \end{bmatrix} \begin{bmatrix} 12 \\ 4 \\ p \end{bmatrix}$$

$$a = \frac{1}{6}(12 - 12 + 2p) = \frac{2p}{6} = \frac{p}{3}$$

$$b = \frac{1}{6}(-36 + 12 + 0) = -4$$

$$c = \frac{1}{6}(24 + 24 - 2p) = \frac{48 - 2p}{6} = 8 - \frac{p}{3}.$$

(d)

For the function to be linear, $a = 0$.

$$\frac{p}{3} = 0 \implies p = 0.$$

If $a = 0$, the curve is a straight line $y = -4x + 8$. Geometrically, this implies that the three points $(-1, 12)$, $(1, 4)$, and $(2, 0)$ are perfectly collinear.**Takeaways 4.1**

- **Vandermonde Matrices:** The coefficient matrix formed by fitting a polynomial to coordinate points is known as a Vandermonde matrix. Its determinant is always non-zero as long as the x -coordinates are distinct, guaranteeing a unique polynomial fit.
- **Parametric Matrix Operations:** Treating unknowns as parameters (like p) within a matrix equation allows you to generalize solutions for an entire family of curves at once, rather than re-solving the system from scratch.
- **Geometric Limits:** Setting the leading coefficient of a polynomial to zero algebraically represents flattening the curve's curvature, visually forcing the defining points into a straight line.

Problem 4.2: Matrix Systems and Intersecting Planes

Consider the following system of homogeneous linear equations, which represent three planes passing through the origin:

$$\begin{aligned}x - y - z &= 0 \\2x + 4y - z &= 0 \\3x + 3y + kz &= 0\end{aligned}$$

where k is a real constant.

- Write the system in the matrix form $AX = O$, where O is the zero vector. Find the determinant of the coefficient matrix A in terms of k , and determine the specific value of k for which A is a singular matrix.
- For the singular value of k found in part (a), show algebraically that the system is consistent. Let $y = -t$ and solve the simultaneous equations for x and z strictly in terms of the parameter t .
- Hence, write the vector equation of the line of intersection of these planes in the form $\mathbf{r} = t(a\mathbf{i} + b\mathbf{j} + c\mathbf{k})$. Briefly explain how this line relates to the intersection of just the first two planes in the system.
- If $k = 4$, determine the unique solution to the system algebraically, and explain what this result means about the geometric arrangement of the three planes.

Hint:

- Part (a):** The coefficient matrix A consists of the multipliers of x , y , and z . Calculate the determinant of this 3×3 matrix by expanding across the top row. A matrix is singular when its determinant is exactly zero.
- Part (b):** Substitute the value of k into the third equation. To prove consistency, look for a linear combination of the first two equations (e.g., adding them together) that perfectly matches the third equation. Then, substitute $y = -t$ into the first two equations and solve the resulting system for x and z .
- Part (c):** Construct a position vector $\mathbf{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$ using your parametric expressions from part (b), and factor out t .
- Part (d):** If $k = 4$, calculate $\det(A)$ to confirm it is non-zero (meaning the matrix is invertible). Since the constant vector is the zero vector, multiplying it by any inverse matrix will always yield the zero vector.

Solution 4.2

(a)

The matrix equation is $\begin{bmatrix} 1 & -1 & -1 \\ 2 & 4 & -1 \\ 3 & 3 & k \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$.

$$\det(A) = 1(4k - (-3)) - (-1)(2k - (-3)) + (-1)(6 - 12)$$

$$\det(A) = (4k + 3) + (2k + 3) - (-6) = 6k + 12.$$

$$\text{For } A \text{ to be singular, } \det(A) = 0 \implies 6k + 12 = 0 \implies k = -2.$$

(b)

Substitute $k = -2$: the third equation is $3x + 3y - 2z = 0$.

Adding Equation 1 and Equation 2 gives:

$$(x - y - z) + (2x + 4y - z) = 3x + 3y - 2z = 0.$$

Since this perfectly matches the third equation, the system is consistent (the third plane is dependent on the first two).

Let $y = -t$.

$$\text{Equation 1 becomes: } x - (-t) - z = 0 \implies x - z = -t.$$

$$\text{Equation 2 becomes: } 2x + 4(-t) - z = 0 \implies 2x - z = 4t.$$

$$\text{Subtracting Equation 1 from Equation 2: } (2x - z) - (x - z) = 4t - (-t) \implies x = 5t.$$

$$\text{Substituting } x \text{ back into Equation 1: } 5t - z = -t \implies z = 6t.$$

The solution is $x = 5t, y = -t, z = 6t$.

(c)

The vector equation is $\mathbf{r} = 5t\mathbf{i} - t\mathbf{j} + 6t\mathbf{k} = t(5\mathbf{i} - \mathbf{j} + 6\mathbf{k})$.

Because the third plane is simply a linear combination of the first two, it contains their line of intersection.

Therefore, this vector equation represents the straight line intersection of the planes $x - y - z = 0$ and $2x + 4y - z = 0$ through the origin.

(d)

$$\text{If } k = 4, \det(A) = 6(4) + 12 = 36.$$

Since $\det(A) \neq 0$, matrix A is invertible.

$$X = A^{-1}O = O \implies x = 0, y = 0, z = 0.$$

Geometrically, this means the three planes intersect at a single, unique point, which is the origin $(0, 0, 0)$.

Takeaways 4.2

- **Homogeneous Systems:** A system where all constants are zero always has at least one solution (the trivial solution: the origin). Whether it has *infinite* solutions depends on singularity.
- **Singularity and Intersection:** If the determinant of a 3×3 coefficient matrix is zero, the planes form a dependent system. If consistent, they intersect along a common line (forming a sheaf of planes).
- **Parametric Conversions:** Expressing an infinite solution set algebraically using a parameter (t) maps directly to the Cartesian components of a 3D vector line equation.

Problem 4.3: Leslie Matrices and Parametric Population Modeling

Based on standard Leslie matrix population modeling principles, consider a modified species of female insect with a maximum life span of 3 months. The population is divided into three age groups of 1 month each.

The Leslie matrix, L , for this population is given by:

$$L = \begin{bmatrix} 0 & k & 4 \\ 0.5 & 0 & 0 \\ 0 & 0.2 & 0 \end{bmatrix}$$

where k represents the unknown fecundity (birth rate) of the second age group. Assume that each of the three age groups initially consists of 400 females, represented by the initial population state matrix P_0 .

- Express the population column matrix after 1 month, P_1 , strictly in terms of k .
- Given that the total population of female insects after 1 month is exactly 2680, algebraically determine the value of k .
- Using your value of k from part (b), an environmental pesticide is introduced that immediately halves the survival rates of all age groups moving forward. Construct the new Leslie matrix, L_{new} , and calculate the exact population matrix for month 2, P_2 .
- Evaluate the determinant of the original matrix L . Deduce whether the original population transition model is mathematically reversible (invertible), and algebraically explain why the value of the determinant is completely independent of the fecundity parameter k .

Hint:

- Part (a):** The initial population matrix is a 3×1 column vector containing the value 400 in each row. Multiply the 3×3 Leslie matrix by this vector using standard row-by-column multiplication.
- Part (b):** Sum the three row elements of your P_1 matrix to find an expression for the total population in terms of k . Set this expression equal to 2680 and solve the resulting linear equation.
- Part (c):** The survival rates are located on the sub-diagonal of the Leslie matrix (the 0.5 and 0.2). Divide these by 2 to create L_{new} . To find P_2 , multiply L_{new} by the numerical P_1 matrix you found in part (b).
- Part (d):** Calculate the determinant of the 3×3 matrix L by expanding along the first column or the third row, as they contain multiple zeros. A matrix model is reversible (has an inverse) if its determinant is non-zero.

Solution 4.3**(a)**

$$P_0 = \begin{bmatrix} 400 \\ 400 \\ 400 \end{bmatrix}$$

$$P_1 = LP_0 = \begin{bmatrix} 0 & k & 4 \\ 0.5 & 0 & 0 \\ 0 & 0.2 & 0 \end{bmatrix} \begin{bmatrix} 400 \\ 400 \\ 400 \end{bmatrix}$$

$$P_1 = \begin{bmatrix} 0(400) + k(400) + 4(400) \\ 0.5(400) + 0 + 0 \\ 0 + 0.2(400) + 0 \end{bmatrix} = \begin{bmatrix} 400k + 1600 \\ 200 \\ 80 \end{bmatrix}.$$

(b)

The total population is the sum of the elements in P_1 :

$$\text{Total} = (400k + 1600) + 200 + 80 = 400k + 1880.$$

We are given that the total population is 2680.

$$400k + 1880 = 2680 \implies 400k = 800 \implies k = 2.$$

(c)

$$\text{With } k = 2, P_1 = \begin{bmatrix} 2400 \\ 200 \\ 80 \end{bmatrix}.$$

The pesticide halves the survival rates ($0.5 \rightarrow 0.25$ and $0.2 \rightarrow 0.1$). The fecundities in the top row remain unchanged.

$$L_{\text{new}} = \begin{bmatrix} 0 & 2 & 4 \\ 0.25 & 0 & 0 \\ 0 & 0.1 & 0 \end{bmatrix}$$

$$P_2 = L_{\text{new}}P_1 = \begin{bmatrix} 0 & 2 & 4 \\ 0.25 & 0 & 0 \\ 0 & 0.1 & 0 \end{bmatrix} \begin{bmatrix} 2400 \\ 200 \\ 80 \end{bmatrix}$$

$$P_2 = \begin{bmatrix} 0 + 2(200) + 4(80) \\ 0.25(2400) + 0 + 0 \\ 0 + 0.1(200) + 0 \end{bmatrix} = \begin{bmatrix} 400 + 320 \\ 600 \\ 20 \end{bmatrix} = \begin{bmatrix} 720 \\ 600 \\ 20 \end{bmatrix}.$$

After 2 months, there are 720 insects in age group 1, 600 in age group 2, and 20 in age group 3.

(d)

Expanding the determinant of L along the first column:

$$\det(L) = 0 - 0.5 \det \begin{bmatrix} k & 4 \\ 0.2 & 0 \end{bmatrix} + 0$$

$$\det(L) = -0.5((k)(0) - (4)(0.2)) = -0.5(0 - 0.8) = 0.4.$$

Since $\det(L) = 0.4 \neq 0$, the matrix L is invertible, meaning the population transition can be mathematically reversed to find past population states.

The value of k disappears during the determinant calculation because expanding the 2×2 minor matrix forces k to be multiplied by 0 (the bottom right element of the Leslie matrix). In a standard Leslie matrix, the determinant depends strictly on the product of the survival rates and the final age group's fecundity.

Takeaways 4.3

- **Parametric Modeling:** Using matrices to solve for unknown biological parameters (like fecundity k) demonstrates how models are calibrated using empirical field data (e.g., total population counts).
- **Dynamic Matrices:** Real-world matrix models are rarely static. Understanding how to manually manipulate specific elements (like survival rates) allows for the simulation of environmental impacts over discrete time steps.
- **Biological Meaning of Determinants:** The algebraic structure of a Leslie matrix dictates that its determinant is independent of intermediate birth rates. A singular Leslie matrix (determinant of zero) only occurs if an age group has a zero survival rate or if the oldest age group becomes sterile, mathematically erasing historical population data.

Problem 4.4: Leslie Matrices and Stable Populations

Consider a certain species of fish divided into three age groups, each one year in length. The fish do not reproduce in their first year, produce an average of 1.5 offspring in their second year, and produce an unknown average of k offspring in their third year, after which they die. Assume that 40% of the fish survive their first year, and 50% of those survivors make it to their third year. The initial population consists of 1000 newborns and zero older fish.

- Construct the Leslie matrix, L , for this fish population and calculate the population state column matrix after 2 years, P_2 .
- Determine an algebraic expression for the number of newborns present in year 3 in terms of k .
- For the population to eventually reach a strict long-term stable size (a long-term growth rate of exactly 1), determine the required exact value of the parameter k .
- Using the value of k determined in part (c), find the long-term stable age distribution of the fish population, expressing the number of fish in each age group as a simplified fraction of the total population.

Hint:

- **Part (a):** The Leslie matrix is a 3×3 matrix where the first row contains the fecundities (birth rates) and the sub-diagonal contains the survival rates. To find P_2 , multiply the initial population column matrix by L twice (or by L^2).
- **Part (b):** Multiply your result for P_2 by the Leslie matrix L to find P_3 . The top element of this resulting matrix represents the newborns.
- **Part (c):** A long-term growth rate of 1 means the population state eventually satisfies the equation $LP = P$, where $P = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$. Expand this matrix equation into a system of simultaneous linear equations and solve for k .
- **Part (d):** Using the system of equations from part (c), express the number of fish in each age group (x , y , and z) in terms of a single variable, such as y . Sum these to find the total population, and then write each age group as a fraction of that total.

Solution 4.4

(a)

The Leslie matrix is $L = \begin{bmatrix} 0 & 1.5 & k \\ 0.4 & 0 & 0 \\ 0 & 0.5 & 0 \end{bmatrix}$.

The initial population is $P_0 = \begin{bmatrix} 1000 \\ 0 \\ 0 \end{bmatrix}$.

$$P_1 = LP_0 = \begin{bmatrix} 0 & 1.5 & k \\ 0.4 & 0 & 0 \\ 0 & 0.5 & 0 \end{bmatrix} \begin{bmatrix} 1000 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 400 \\ 0 \end{bmatrix}.$$

$$P_2 = LP_1 = \begin{bmatrix} 0 & 1.5 & k \\ 0.4 & 0 & 0 \\ 0 & 0.5 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 400 \\ 0 \end{bmatrix} = \begin{bmatrix} 600 \\ 0 \\ 200 \end{bmatrix}.$$

(b)

$$P_3 = LP_2 = \begin{bmatrix} 0 & 1.5 & k \\ 0.4 & 0 & 0 \\ 0 & 0.5 & 0 \end{bmatrix} \begin{bmatrix} 600 \\ 0 \\ 200 \end{bmatrix} = \begin{bmatrix} 0(600) + 1.5(0) + k(200) \\ 0.4(600) + 0 + 0 \\ 0 + 0.5(0) + 0 \end{bmatrix} = \begin{bmatrix} 200k \\ 240 \\ 0 \end{bmatrix}.$$

The number of newborns in year 3 is $200k$.

(c)

For a long-term growth rate of 1, the stable state satisfies $LP = P$:

$$\begin{bmatrix} 0 & 1.5 & k \\ 0.4 & 0 & 0 \\ 0 & 0.5 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} x \\ y \\ z \end{bmatrix}$$

This yields the system of equations:

$$(1) \quad 1.5y + kz = x$$

$$(2) \quad 0.4x = y \implies x = 2.5y$$

$$(3) \quad 0.5y = z$$

Substitute (2) and (3) into (1):

$$1.5y + k(0.5y) = 2.5y$$

Assuming a non-zero population, divide by y :

$$1.5 + 0.5k = 2.5 \implies 0.5k = 1 \implies k = 2.$$

(d)

With $k = 2$, the age group relations are $x = 2.5y$ and $z = 0.5y$.

Total population $T = x + y + z = 2.5y + y + 0.5y = 4y$.

Proportion of age group 1: $\frac{x}{T} = \frac{2.5y}{4y} = \frac{2.5}{4} = \frac{5}{8}$.

Proportion of age group 2: $\frac{y}{T} = \frac{1y}{4y} = \frac{1}{4} = \frac{2}{8}$.

Proportion of age group 3: $\frac{z}{T} = \frac{0.5y}{4y} = \frac{0.5}{4} = \frac{1}{8}$.

The stable age distribution is $\frac{5}{8}$ newborns, $\frac{1}{4}$ one-year-olds, and $\frac{1}{8}$ two-year-olds.

Takeaways 4.4

- **Eigenvalues in Context:** A long-term growth rate of 1 implies that the dominant eigenvalue of the Leslie matrix is $\lambda = 1$. Setting $LP = 1P$ allows you to bypass the characteristic polynomial and directly solve for matrix parameters.
- **Stable Age Distributions:** Once a population reaches its long-term growth trajectory (determined by the dominant eigenvalue), the *ratio* of the age groups remains constant year over year, representing the corresponding eigenvector.
- **Matrix Generality:** Using variables within matrices (like k) is a standard QCAA technique to test algebraic manipulation and conceptual understanding of matrix operations simultaneously.

Problem 4.5: Leslie Matrices and Periodic Populations

A particular species of insect has a maximum life span of 3 months. The female insects reproduce only in their third month of life. When the female population is divided into three age groups, each of length one month, the Leslie matrix takes the form:

$$L = \begin{bmatrix} 0 & 0 & 8 \\ 0.5 & 0 & 0 \\ 0 & 0.25 & 0 \end{bmatrix}$$

- (a) Evaluate L^3 . Hence, deduce the long-term growth rate of this population and describe the cyclical nature of the population sizes every 3 months.
- (b) Determine the stable age distribution for this population by finding the vector $\mathbf{v} = \begin{bmatrix} 1 \\ c \\ d \end{bmatrix}$ such that $L\mathbf{v} = \mathbf{v}$. Express this stable distribution as fractions of the total population.

- (c) Suppose the initial female population consists purely of newborns: $P_0 = \begin{bmatrix} 1200 \\ 0 \\ 0 \end{bmatrix}$.

Calculate the population matrices P_1 , P_2 , and P_3 . Explain why this specific population does not appear to exhibit a stable age distribution over these initial months.

- (d) To stabilize the population immediately, a conservationist proposes adding a specific number of 2nd-month and 3rd-month insects to the initial population P_0 . Determine the exact number of insects that must be added to age groups 2 and 3 so that the new initial population P_{new} satisfies $LP_{\text{new}} = P_{\text{new}}$.

Hint:

- **Part (a):** Multiply L by itself to find L^2 , and then multiply by L again to find L^3 . Think about what $P_3 = L^3P_0$ means if L^3 simplifies to a scalar multiple of the identity matrix.
- **Part (b):** Multiply matrix L by the column vector $\mathbf{v}$ and set the result equal to $\mathbf{v}$. This will give you a system of linear equations to solve for c and d . Sum the components to find the fractional distribution.
- **Part (c):** Use the recursive formula $P_n = LP_{n-1}$. Observe where the non-zero values shift each month.
- **Part (d):** For P_{new} to be perfectly stable, it must be a scalar multiple of the stable age vector $\mathbf{v}$ you found in part (b). Since the number of newborns is fixed at 1200, use this to find the scalar multiplier.

Solution 4.5**(a)**

$$L^2 = \begin{bmatrix} 0 & 0 & 8 \\ 0.5 & 0 & 0 \\ 0 & 0.25 & 0 \end{bmatrix} \begin{bmatrix} 0 & 0 & 8 \\ 0.5 & 0 & 0 \\ 0 & 0.25 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 2 & 0 \\ 0 & 0 & 4 \\ 0.125 & 0 & 0 \end{bmatrix}.$$

$$L^3 = L^2 L = \begin{bmatrix} 0 & 2 & 0 \\ 0 & 0 & 4 \\ 0.125 & 0 & 0 \end{bmatrix} \begin{bmatrix} 0 & 0 & 8 \\ 0.5 & 0 & 0 \\ 0 & 0.25 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I.$$

Since $L^3 = I$, applying the matrix 3 times leaves the population unchanged ($P_3 = IP_0 = P_0$). The long-term growth rate is 1. The population perfectly cycles every 3 months.

(b)Setting $L\mathbf{v} = \mathbf{v}$:

$$\begin{bmatrix} 0 & 0 & 8 \\ 0.5 & 0 & 0 \\ 0 & 0.25 & 0 \end{bmatrix} \begin{bmatrix} 1 \\ c \\ d \end{bmatrix} = \begin{bmatrix} 1 \\ c \\ d \end{bmatrix} \implies \begin{bmatrix} 8d \\ 0.5 \\ 0.25c \end{bmatrix} = \begin{bmatrix} 1 \\ c \\ d \end{bmatrix}.$$

From row 2: $c = 0.5 = 1/2$.From row 1: $8d = 1 \implies d = 1/8$.

The vector is $\begin{bmatrix} 1 \\ 1/2 \\ 1/8 \end{bmatrix}$. The total parts are $1 + \frac{1}{2} + \frac{1}{8} = \frac{13}{8}$.

Fractions: Age 1 is $\frac{1}{13/8} = \frac{8}{13}$, Age 2 is $\frac{1/2}{13/8} = \frac{4}{13}$, Age 3 is $\frac{1/8}{13/8} = \frac{1}{13}$.

(c)

$$P_1 = LP_0 = \begin{bmatrix} 0 & 0 & 8 \\ 0.5 & 0 & 0 \\ 0 & 0.25 & 0 \end{bmatrix} \begin{bmatrix} 1200 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 600 \\ 0 \end{bmatrix}.$$

$$P_2 = LP_1 = \begin{bmatrix} 0 & 0 & 8 \\ 0.5 & 0 & 0 \\ 0 & 0.25 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 600 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \\ 150 \end{bmatrix}.$$

$$P_3 = LP_2 = \begin{bmatrix} 0 & 0 & 8 \\ 0.5 & 0 & 0 \\ 0 & 0.25 & 0 \end{bmatrix} \begin{bmatrix} 0 \\ 0 \\ 150 \end{bmatrix} = \begin{bmatrix} 1200 \\ 0 \\ 0 \end{bmatrix}.$$

The population is not in a stable age distribution because the initial state consisted only of one age group. In a species that only reproduces in its final life stage, a cohort gap creates a traveling demographic wave that never smooths out unless multiple generations overlap initially.

(d)

We need $P_{\text{new}} = \begin{bmatrix} 1200 \\ y \\ z \end{bmatrix}$ to be a scalar multiple of $\mathbf{v} = \begin{bmatrix} 1 \\ 1/2 \\ 1/8 \end{bmatrix}$.

The scalar multiplier is 1200 (since $1 \times 1200 = 1200$).

$$y = 1200 \times \frac{1}{2} = 600.$$

$$z = 1200 \times \frac{1}{8} = 150.$$

Therefore, the conservationist must add 600 2nd-month insects and 150 3rd-month insects.

Takeaways 4.5

- **Imprimitive Matrices:** Leslie matrices where reproduction only occurs in a single, final age group are “imprimitive.” They do not naturally converge to a stable age distribution from an arbitrary starting point; instead, they oscillate cyclically.
- **Eigenvectors as Proportions:** The steady-state vector (corresponding to a growth multiplier of 1) represents the exact population ratio required to maintain a static, unchanging demographic structure year over year.
- **Matrix Exponents:** Discovering that $L^k = I$ is a powerful algebraic shortcut. It immediately proves that the system is periodic with period k and that the long-term growth multiplier is exactly 1.

4.3 Challenge Corner

Problem 4.6: Parametric Simultaneous Equations and Matrices

Consider the simultaneous linear equations involving real constants m and n :

$$mx + 2y = 4$$

$$8x + my = n$$

- Express this system as a matrix equation of the form $AX = B$. Hence, determine the inverse matrix A^{-1} in terms of m , stating the specific values of m for which A^{-1} is undefined.
- Determine the values of m for which the system does *not* have a unique solution.
- Given that m takes the positive value found in part (b), determine the exact value of n that results in an infinite set of solutions. Furthermore, provide a geometric interpretation of the system if $n = 10$ instead.
- Suppose $n = 2m$. Assuming a unique solution exists, use the matrix inverse method to solve for x and y in terms of m , expressing your final answers in their simplest form.

Hint:

- **Part (a):** Set up the coefficients of x and y into a 2×2 matrix. The formula for the inverse includes dividing by the determinant, $ad - bc$. The inverse is undefined when this determinant is equal to zero.
- **Part (b):** A unique solution exists if and only if the coefficient matrix A is invertible. Therefore, a unique solution does *not* exist when the determinant you found in part (a) is zero.
- **Part (c):** Substitute the positive value of m into the original equations. For infinite solutions, the two line equations must be mathematically equivalent (one is a scalar multiple of the other). If the lines have the same gradient but different y -intercepts, what does that mean geometrically?
- **Part (d):** Substitute $n = 2m$ into your column matrix B . Multiply A^{-1} by B using standard row-by-column multiplication. Simplify the resulting algebraic expressions for x and y by factoring out common terms.

Solution 4.6**(a)**

Matrix equation: $\begin{bmatrix} m & 2 \\ 8 & m \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 4 \\ n \end{bmatrix}$.

The determinant is $\det(A) = (m)(m) - (2)(8) = m^2 - 16$.

$$A^{-1} = \frac{1}{m^2 - 16} \begin{bmatrix} m & -2 \\ -8 & m \end{bmatrix}.$$

The inverse is undefined when $m^2 - 16 = 0$, which occurs when $m = 4$ or $m = -4$.

(b)

The system does not have a unique solution when the coefficient matrix is singular (i.e., $\det(A) = 0$).

From part (a), this occurs when $m = 4$ or $m = -4$.

(c)

Let $m = 4$. The system becomes:

$$(1) \quad 4x + 2y = 4 \implies 2x + y = 2$$

$$(2) \quad 8x + 4y = n \implies 2x + y = \frac{n}{4}$$

For an infinite set of solutions, the equations must represent the same line. Therefore, $2 = \frac{n}{4} \implies n = 8$.

If $n = 10$, the equations are $2x + y = 2$ and $2x + y = 2.5$. These equations represent distinct, parallel lines that never intersect, meaning there is no solution.

(d)

Let $n = 2m$. The matrix B becomes $\begin{bmatrix} 4 \\ 2m \end{bmatrix}$.

$$X = A^{-1}B = \frac{1}{m^2 - 16} \begin{bmatrix} m & -2 \\ -8 & m \end{bmatrix} \begin{bmatrix} 4 \\ 2m \end{bmatrix}$$

$$X = \frac{1}{m^2 - 16} \begin{bmatrix} (m)(4) + (-2)(2m) \\ (-8)(4) + (m)(2m) \end{bmatrix}$$

$$X = \frac{1}{m^2 - 16} \begin{bmatrix} 4m - 4m \\ -32 + 2m^2 \end{bmatrix} = \frac{1}{m^2 - 16} \begin{bmatrix} 0 \\ 2(m^2 - 16) \end{bmatrix}$$

$$X = \begin{bmatrix} 0 \\ 2 \end{bmatrix}.$$

The unique solution simplifies elegantly to $x = 0$ and $y = 2$ (for $m \neq \pm 4$).

Takeaways 4.6

- **Parametric Determinants:** Using parameters in the coefficient matrix directly links algebraic singularity (determinant = 0) to geometric relationships (parallel or coincident lines).
- **Infinite vs. No Solution:** Once a matrix is singular, comparing the constant terms (the B matrix) dictates whether the lines are coincident (infinite solutions) or strictly parallel (no solution).
- **Power of Matrix Algebra:** Part (d) demonstrates how matrix inverse multiplication can elegantly simplify complex parametric systems into clean integer solutions through algebraic factorization.

Problem 4.7: Matrix Systems and Geometric Transformations

Consider the following system of linear equations:

$$\begin{aligned}\sqrt{3}x - y &= 4\sqrt{3} \\ x + \sqrt{3}y &= 4\end{aligned}$$

- Write this system in the matrix form $AX = B$, and determine the exact value of $\det(A)$.
- Determine A^{-1} , and hence solve the system of equations algebraically for $X = \begin{bmatrix} x \\ y \end{bmatrix}$.
- Express the coefficient matrix A in the form $k \begin{bmatrix} \cos \theta & -\sin \theta \\ \sin \theta & \cos \theta \end{bmatrix}$, where $k > 0$ and $0 \leq \theta \leq \frac{\pi}{2}$. Hence, describe the two sequential geometric transformations represented by the matrix A .
- The matrix equation $AX = B$ can be interpreted geometrically as finding a pre-image point (x, y) that maps to the image point $(4\sqrt{3}, 4)$ under transformation A . Using your geometric description from part (c), deduce the coordinates of (x, y) using polar coordinates and transformations, verifying your algebraic answer from part (b).

- Hint:**
- **Part (a):** Extract the coefficients of x and y carefully. The determinant is calculated as $ad - bc$. Be mindful of squaring surds and negative signs.
 - **Part (b):** The inverse of a 2×2 matrix is $\frac{1}{\det(A)} \begin{bmatrix} d & -c \\ -b & a \end{bmatrix}$. To solve for X , pre-multiply the constant matrix B by A^{-1} .
 - **Part (c):** Notice that the determinant of a pure rotation matrix is always 1. The determinant of A is not 1, which implies a dilation is present. The dilation factor k is equal to $\sqrt{\det(A)}$. Factor this k out of matrix A to reveal the standard rotation matrix.
 - **Part (d):** Convert the image point $(4\sqrt{3}, 4)$ into polar form (r, ϕ) . Since applying A dilates and rotates the pre-image *forward*, applying A^{-1} to find the pre-image must do the exact opposite: divide the radius by k and rotate *backwards* by θ .

Solution 4.7

(a)

Matrix form: $\begin{bmatrix} \sqrt{3} & -1 \\ 1 & \sqrt{3} \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} 4\sqrt{3} \\ 4 \end{bmatrix}$
 $\det(A) = (\sqrt{3})(\sqrt{3}) - (-1)(1) = 3 - (-1) = 4.$

(b)

$$A^{-1} = \frac{1}{4} \begin{bmatrix} \sqrt{3} & 1 \\ -1 & \sqrt{3} \end{bmatrix}.$$

$$X = A^{-1}B = \frac{1}{4} \begin{bmatrix} \sqrt{3} & 1 \\ -1 & \sqrt{3} \end{bmatrix} \begin{bmatrix} 4\sqrt{3} \\ 4 \end{bmatrix}$$

$$X = \frac{1}{4} \begin{bmatrix} \sqrt{3}(4\sqrt{3}) + 1(4) \\ -1(4\sqrt{3}) + \sqrt{3}(4) \end{bmatrix} = \frac{1}{4} \begin{bmatrix} 12 + 4 \\ -4\sqrt{3} + 4\sqrt{3} \end{bmatrix} = \frac{1}{4} \begin{bmatrix} 16 \\ 0 \end{bmatrix} = \begin{bmatrix} 4 \\ 0 \end{bmatrix}.$$

Therefore, $x = 4$ and $y = 0$.

(c)

Since $\det(A) = 4$, the dilation factor is $k = \sqrt{4} = 2$.

Factoring out 2: $A = 2 \begin{bmatrix} \frac{\sqrt{3}}{2} & -\frac{1}{2} \\ \frac{1}{2} & \frac{\sqrt{3}}{2} \end{bmatrix}.$

We need $\cos \theta = \frac{\sqrt{3}}{2}$ and $\sin \theta = \frac{1}{2}$, which gives $\theta = \frac{\pi}{6}$ (or 30°).

Thus, $A = 2 \begin{bmatrix} \cos(\frac{\pi}{6}) & -\sin(\frac{\pi}{6}) \\ \sin(\frac{\pi}{6}) & \cos(\frac{\pi}{6}) \end{bmatrix}.$

Geometrically, matrix A represents a **dilation by a factor of 2** from the origin, followed by an **anti-clockwise rotation of $\frac{\pi}{6}$ radians** about the origin.

(d)

The image point is $(4\sqrt{3}, 4)$. In polar coordinates:

$$r = \sqrt{(4\sqrt{3})^2 + 4^2} = \sqrt{48 + 16} = \sqrt{64} = 8.$$

$$\text{Angle } \phi = \tan^{-1} \left(\frac{4}{4\sqrt{3}} \right) = \tan^{-1} \left(\frac{1}{\sqrt{3}} \right) = \frac{\pi}{6}.$$

So the image point is $(8, \frac{\pi}{6})$ in polar.

The inverse transformation A^{-1} reverses the effects of A : it dilates by $\frac{1}{2}$ and rotates clockwise by $\frac{\pi}{6}$.

New radius $r' = 8 \times \frac{1}{2} = 4$.

New angle $\phi' = \frac{\pi}{6} - \frac{\pi}{6} = 0$.

The pre-image in polar coordinates is $(4, 0)$, which corresponds to the Cartesian coordinates $(4, 0)$. This verifies the algebraic solution $x = 4, y = 0$.

Takeaways 4.7

- **Algebra Meets Geometry:** Solving a system of linear equations is algebraically identical to finding the pre-image of a point under a 2D matrix transformation.
- **Identifying Transformations:** Any matrix of the form $\begin{bmatrix} a & -b \\ b & a \end{bmatrix}$ represents a combined rotation and dilation. The dilation factor is $\sqrt{a^2 + b^2}$ (or $\sqrt{\det(A)}$), and the rotation angle θ is found using right-triangle trigonometry with a and b .
- **Inverse Transformations:** If matrix A scales by k and rotates by θ , its inverse A^{-1} scales by $\frac{1}{k}$ and rotates by $-\theta$.

Problem 4.8: Matrix Determinants and System Solutions

Consider the following system of simultaneous linear equations involving the real parameters m and p :

$$\begin{aligned} mx + 2y &= p \\ 12x + (m - 5)y &= 6 \end{aligned}$$

- Express this system in the matrix form $AX = B$. Determine the values of m for which matrix A is not an invertible matrix.
- Let m take the positive value found in part (a). Determine the exact value of p required for the system to have an infinite number of solution pairs. Describe the geometric relationship between the two linear equations in this specific case.
- Let m take the negative value found in part (a). Determine the restriction on p such that the system of equations has no solutions. Describe the geometric relationship between the two linear equations in this case.
- Suppose $p = 4$, and assume that m is chosen such that a unique solution exists. Use the matrix inverse method to solve for the x -coordinate of the solution in terms of m , expressing your final answer as a simplified rational fraction.

- Hint:**
- **Part (a):** A matrix is not invertible (it is singular) when its determinant equals zero. Extract the coefficients, calculate the determinant using $ad - bc$, set it to zero, and factor the resulting quadratic equation.
 - **Part (b):** Substitute your positive m value into the original system. For a linear system to have infinite solutions, the two equations must be scalar multiples of one another (representing coincident lines).
 - **Part (c):** Substitute your negative m value into the original system. For a linear system to have no solutions, the lines must have identical gradients but different y -intercepts. What value must p not equal to prevent the lines from overlapping?
 - **Part (d):** The inverse matrix is $A^{-1} = \frac{1}{\det(A)} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$. To find x , you only need to calculate the top row of the matrix multiplication $A^{-1}B$. Look for common factors in the numerator and denominator to simplify.

Solution 4.8

(a)

Matrix equation: $\begin{bmatrix} m & 2 \\ 12 & m-5 \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} p \\ 6 \end{bmatrix}$.

Matrix A is not invertible when $\det(A) = 0$.

$$\det(A) = (m)(m-5) - (2)(12) = m^2 - 5m - 24.$$

$$\text{Setting to zero: } m^2 - 5m - 24 = 0 \implies (m-8)(m+3) = 0.$$

Therefore, A is singular when $m = 8$ or $m = -3$.

(b)

Let $m = 8$. The system becomes:

$$(1) \ 8x + 2y = p \implies 4x + y = \frac{p}{2}$$

$$(2) \ 12x + 3y = 6 \implies 4x + y = 2$$

For infinite solutions, the equations must be equivalent.

$$\text{Therefore, } \frac{p}{2} = 2 \implies p = 4.$$

Geometrically, the equations represent **coincident (identical) straight lines**.

(c)

Let $m = -3$. The system becomes:

$$(1) \ -3x + 2y = p$$

$$(2) \ 12x - 8y = 6 \implies -3x + 2y = -1.5$$

For the system to have no solutions, the equations must be mathematically inconsistent.

Therefore, p can be any real number except -1.5 (i.e., $p \neq -1.5$).

Geometrically, the equations represent **distinct parallel lines** that never intersect.

(d)

Let $p = 4$. For a unique solution, $\det(A) = (m-8)(m+3) \neq 0$.

$$X = A^{-1}B = \frac{1}{(m-8)(m+3)} \begin{bmatrix} m-5 & -2 \\ -12 & m \end{bmatrix} \begin{bmatrix} 4 \\ 6 \end{bmatrix}.$$

We only need to evaluate the top row for x :

$$x = \frac{4(m-5)+6(-2)}{(m-8)(m+3)} = \frac{4m-20-12}{(m-8)(m+3)} = \frac{4m-32}{(m-8)(m+3)}.$$

Factoring the numerator:

$$x = \frac{4(m-8)}{(m-8)(m+3)}.$$

Since $m \neq 8$, we can cancel the common factor, yielding the simplified solution:

$$x = \frac{4}{m+3}.$$

Takeaways 4.8

- **The Zero Determinant:** When $\det(A) = 0$, a unique solution does not exist. This algebraically signals that the lines are either parallel or coincident.
- **The Role of Constants:** Once singularity is established by the coefficient matrix A , the constant matrix B dictates whether the system collapses into an infinite solution set or an empty set (no solutions).
- **Parametric Simplification:** In matrix algebra, finding a unique parametric solution often leads to rational expressions where the roots of the determinant (e.g., $m - 8$) appear as cancellable factors in the numerators.

Problem 4.9: Matrix Systems and Parametric Constants

Consider the following system of linear equations involving a real constant k :

$$kx + 2y = k + 1$$

$$8x + ky = 6$$

- Write this system in the matrix form, as $AX = K$. Determine the values of k for which the coefficient matrix A is not an invertible matrix.
- For the negative value of k found in part (a), determine the geometric relationship between the two linear equations. How many pairs does the solution set contain?
- For the positive value of k found in part (a), demonstrate algebraically whether any solutions can be found for this system of equations. Interpret this result geometrically.
- Let $k = 2$. Calculate A^{-1} and hence find the unique solution pair (x, y) that satisfies the system.

Hint:

- **Part (a):** Matrix A contains the coefficients of x and y . It is not invertible (singular) when its determinant, calculated as $ad - bc$, is equal to zero.
- **Part (b):** Substitute the negative k value back into the original equations. If one equation simplifies to become a direct multiple of the other, they represent the exact same line.
- **Part (c):** Substitute the positive k value. Try to solve the system using elimination; you will reach a false mathematical statement, indicating parallel lines that never cross.
- **Part (d):** Use the inverse matrix formula $A^{-1} = \frac{1}{\det(A)} \begin{bmatrix} d & -b \\ -c & a \end{bmatrix}$ and multiply it by the constant column matrix K .

Solution 4.9**(a)**

Matrix form: $\begin{bmatrix} k & 2 \\ 8 & k \end{bmatrix} \begin{bmatrix} x \\ y \end{bmatrix} = \begin{bmatrix} k+1 \\ 6 \end{bmatrix}.$

The matrix A is not invertible when $\det(A) = 0$.

$$\det(A) = (k)(k) - (2)(8) = k^2 - 16.$$

$$\text{Setting to zero: } k^2 - 16 = 0 \implies k = 4 \text{ or } k = -4.$$

(b)

Let $k = -4$. The system becomes:

$$(1) -4x + 2y = -3$$

$$(2) 8x - 4y = 6 \implies -4x + 2y = -3$$

The equations are mathematically identical. Geometrically, they represent **coincident lines**. Because they overlap entirely, the solution set contains an **infinite number of pairs**.

(c)

Let $k = 4$. The system becomes:

$$(1) 4x + 2y = 5$$

$$(2) 8x + 4y = 6 \implies 4x + 2y = 3$$

Subtracting the simplified second equation from the first gives $0 = 2$, which is a contradiction. Therefore, **no solutions can be found**. Geometrically, they represent **distinct parallel lines** that never intersect.

(d)

For $k = 2$, $A = \begin{bmatrix} 2 & 2 \\ 8 & 2 \end{bmatrix}$ and $K = \begin{bmatrix} 3 \\ 6 \end{bmatrix}.$

$$\det(A) = (2)(2) - (2)(8) = 4 - 16 = -12.$$

$$A^{-1} = -\frac{1}{12} \begin{bmatrix} 2 & -2 \\ -8 & 2 \end{bmatrix}.$$

$$X = A^{-1}K = -\frac{1}{12} \begin{bmatrix} 2 & -2 \\ -8 & 2 \end{bmatrix} \begin{bmatrix} 3 \\ 6 \end{bmatrix} = -\frac{1}{12} \begin{bmatrix} 6 - 12 \\ -24 + 12 \end{bmatrix} = -\frac{1}{12} \begin{bmatrix} -6 \\ -12 \end{bmatrix} = \begin{bmatrix} \frac{1}{2} \\ 1 \end{bmatrix}.$$

The unique solution pair is $(\frac{1}{2}, 1)$.

Takeaways 4.9

- **Singularity and Geometry:** A non-invertible (singular) coefficient matrix indicates that the system does not have a unique solution; the lines are either parallel or coincident.
- **The Role of Constant Matrices:** The constant terms determine the final state of a singular system. Equivalent ratios across all terms yield infinite solutions (coincident), while differing constant ratios yield zero solutions (parallel).
- **Direct Solutions:** The matrix inverse method provides a direct, algorithmic approach to solving uniquely determined systems without requiring manual substitution or elimination.

Problem 4.10: Matrix Algebra and Transformations

Consider the 3×3 matrix $A = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 1 & \sqrt{3} \\ 0 & \sqrt{3} & -1 \end{bmatrix}$.

- Determine A^2 and express your result in the form kI , where I is the 3×3 identity matrix and k is a real constant.
- Hence, algebraically deduce the inverse matrix A^{-1} without calculating the determinant.
- A 3D linear transformation represented by matrix A maps a coordinate point $P(x, y, z)$ to the image point $P'(4, -2, 2\sqrt{3})$. Use your result from part (b) to determine the exact coordinates of the original point P .
- Using the properties established in part (a), evaluate A^3 and hence determine an expression for A^5 strictly in terms of A .

Hint:

- **Part (a):** Perform standard 3×3 matrix multiplication ($A \times A$). Be careful when multiplying the surds in the second and third rows.
- **Part (b):** The definition of an inverse matrix is $AA^{-1} = I$. Take your equation from part (a), $A^2 = kI$, and algebraically manipulate it to isolate I on one side of the equation.
- **Part (c):** The transformation is defined by the equation $AP = P'$. To solve for the pre-image P , you must pre-multiply both sides of the equation by A^{-1} .
- **Part (d):** Do not calculate the matrices manually. Use exponent rules for matrices: $A^3 = A^2 \times A$. Substitute the identity relationship you found in part (a) to simplify the powers.

Solution 4.10**(a)**

$$A^2 = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 1 & \sqrt{3} \\ 0 & \sqrt{3} & -1 \end{bmatrix} \begin{bmatrix} 2 & 0 & 0 \\ 0 & 1 & \sqrt{3} \\ 0 & \sqrt{3} & -1 \end{bmatrix}$$

$$A^2 = \begin{bmatrix} (2)(2) + 0 + 0 & 0 + 0 + 0 & 0 + 0 + 0 \\ 0 + 0 + 0 & (1)(1) + (\sqrt{3})(\sqrt{3}) + 0 & (1)(\sqrt{3}) + (\sqrt{3})(-1) + 0 \\ 0 + 0 + 0 & (\sqrt{3})(1) + (-1)(\sqrt{3}) + 0 & (\sqrt{3})(\sqrt{3}) + (-1)(-1) + 0 \end{bmatrix}$$

$$A^2 = \begin{bmatrix} 4 & 0 & 0 \\ 0 & 1 + 3 & \sqrt{3} - \sqrt{3} \\ 0 & \sqrt{3} - \sqrt{3} & 3 + 1 \end{bmatrix} = \begin{bmatrix} 4 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 4 \end{bmatrix}$$

$$A^2 = 4 \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = 4I.$$

Therefore, $k = 4$.**(b)**From part (a), $A^2 = 4I$.

$$A \cdot A = 4I$$

Dividing both sides by the scalar 4:

$$A \cdot \left(\frac{1}{4}A\right) = I$$

Since $A \cdot A^{-1} = I$, it follows that $A^{-1} = \frac{1}{4}A$.

$$A^{-1} = \begin{bmatrix} 1/2 & 0 & 0 \\ 0 & 1/4 & \sqrt{3}/4 \\ 0 & \sqrt{3}/4 & -1/4 \end{bmatrix}.$$

(c)Given $AP = P'$, we have $P = A^{-1}P'$.Using $A^{-1} = \frac{1}{4}A$:

$$P = \frac{1}{4} \begin{bmatrix} 2 & 0 & 0 \\ 0 & 1 & \sqrt{3} \\ 0 & \sqrt{3} & -1 \end{bmatrix} \begin{bmatrix} 4 \\ -2 \\ 2\sqrt{3} \end{bmatrix}$$

$$P = \frac{1}{4} \begin{bmatrix} (2)(4) + 0 + 0 \\ 0 + (1)(-2) + (\sqrt{3})(2\sqrt{3}) \\ 0 + (\sqrt{3})(-2) + (-1)(2\sqrt{3}) \end{bmatrix}$$

$$P = \frac{1}{4} \begin{bmatrix} 8 \\ -2 + 6 \\ -2\sqrt{3} - 2\sqrt{3} \end{bmatrix} = \frac{1}{4} \begin{bmatrix} 8 \\ 4 \\ -4\sqrt{3} \end{bmatrix} = \begin{bmatrix} 2 \\ 1 \\ -\sqrt{3} \end{bmatrix}.$$

The coordinates of P are $(2, 1, -\sqrt{3})$.**(d)**

$$A^3 = A^2 \cdot A = (4I)A = 4A.$$

$$A^5 = A^3 \cdot A^2 = (4A)(4I) = 16A.$$

$$\text{Alternatively, } A^5 = A(A^2)^2 = A(4I)^2 = 16A.$$

Takeaways 4.10

- **The “Hence” Command:** In mathematics, the word “hence” dictates that you must use the result of the previous section. Finding the inverse via $A^2 = kI$ is often faster and less prone to arithmetic error than manually calculating a 3×3 determinant and adjugate matrix.
- **Matrix Power Cycles:** Matrices that satisfy equations like $A^2 = kI$ act similarly to cyclic numbers in modular arithmetic. High powers of the matrix can be drastically simplified without performing tedious sequential multiplication.
- **Geometric Inverses:** Algebraically isolating a coordinate vector by multiplying by an inverse matrix geometrically represents reversing the original 3D transformation to find the starting position.

Problem 4.11: Systems of Linear Equations and 3D Geometry

Consider the following system of simultaneous linear equations representing three planes in 3D space, where k is a real constant:

$$x + 2y - 3z = 4$$

$$x + y + z = 6$$

$$2x + 3y + kz = 10$$

- Express this system in the matrix form $AX = B$. Calculate the determinant of the 3×3 coefficient matrix A in terms of k , and hence determine the value of k for which the matrix is singular.
- For the value of k found in part (a), demonstrate algebraically that the system is consistent (has solutions) rather than inconsistent (no solutions).
- Given that the system is consistent and has infinitely many solutions, let $z = \lambda$. Solve the equations to express x and y in terms of the parameter λ .
- Express the general solution from part (c) as a vector equation of a line in the form $\mathbf{r} = \mathbf{a} + \lambda \mathbf{d}$, and briefly describe the geometric arrangement of the three planes.

Hint:

- **Part (a):** The determinant of a 3×3 matrix can be found by expanding across the top row. A matrix is singular when its determinant is equal to zero.
- **Part (b):** Look for a linear combination of the first two equations that perfectly yields the third equation. If the left-hand side and right-hand side both match, the system is consistent.
- **Part (c):** Subtract the second equation from the first to eliminate x and find y in terms of z . Then substitute this back into the second equation to find x . Finally, replace z with λ .
- **Part (d):** Separate the constant terms from the terms containing λ to form your position vector $\mathbf{a}$ and direction vector $\mathbf{d}$.

Solution 4.11**(a)**

Matrix form: $\begin{bmatrix} 1 & 2 & -3 \\ 1 & 1 & 1 \\ 2 & 3 & k \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 4 \\ 6 \\ 10 \end{bmatrix}.$

$$\det(A) = 1(1(k) - 1(3)) - 2(1(k) - 1(2)) + (-3)(1(3) - 1(2))$$

$$\det(A) = (k - 3) - 2(k - 2) - 3(1)$$

$$\det(A) = k - 3 - 2k + 4 - 3 = -k - 2.$$

$$\text{For } A \text{ to be singular, } \det(A) = 0 \implies -k - 2 = 0 \implies k = -2.$$

(b)

Substitute $k = -2$ into the third equation: $2x + 3y - 2z = 10$.

Let Equation 1 be $x + 2y - 3z = 4$ and Equation 2 be $x + y + z = 6$.

Adding Equation 1 and Equation 2 gives:

$$(x + x) + (2y + y) + (-3z + z) = 4 + 6$$

$$2x + 3y - 2z = 10.$$

Since this perfectly matches the third equation, the third plane contains the line of intersection of the first two planes. The system is consistent.

(c)

Let $z = \lambda$.

Subtracting Eq 2 from Eq 1:

$$(x + 2y - 3z) - (x + y + z) = 4 - 6$$

$$y - 4z = -2 \implies y = 4z - 2 \implies y = 4\lambda - 2.$$

Substitute $y = 4\lambda - 2$ into Eq 2:

$$x + (4\lambda - 2) + \lambda = 6$$

$$x + 5\lambda - 2 = 6 \implies x = 8 - 5\lambda.$$

(d)

From part (c), we have $x = 8 - 5\lambda$, $y = -2 + 4\lambda$, $z = 0 + 1\lambda$.

In vector form:

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 8 \\ -2 \\ 0 \end{bmatrix} + \lambda \begin{bmatrix} -5 \\ 4 \\ 1 \end{bmatrix}.$$

Geometrically, the three planes intersect along a single common line (forming a sheaf of planes).

Takeaways 4.11

- **Underdetermined vs. Singular Systems:** The original 2×3 system implicitly represented infinite solutions because there were more variables than equations. By adding a dependent third equation, we transition to a 3×3 singular matrix scenario, a core concept in QCAA Specialist Mathematics.
- **Singularity and Consistency:** A determinant of zero only tells you a unique solution does not exist. You must verify algebraically whether the planes are parallel/form a prism (inconsistent) or intersect along a line (consistent).
- **Parametric Lines:** The process of setting a free variable to λ is exactly how the Cartesian equations of intersecting planes are converted into the vector equation of a 3D line.

Problem 4.12: Augmented Matrices and Parametric Solutions

Consider the following system of linear equations involving four variables (x, y, z, w) and a real constant k :

$$\begin{aligned}x + 2y - z + w &= 5 \\2x - y + 3z - 2w &= 0 \\3x + y + 2z - w &= k\end{aligned}$$

- Set up the augmented matrix for this system. By applying elementary row operations (specifically, examining $R_1 + R_2$), determine the exact value of k for which this system is consistent (has solutions).
- Given that the system is consistent, it will have a doubly-infinite family of solutions. Let $z = \lambda$ and $w = \mu$. Express x and y strictly in terms of the parameters λ and μ .
- A new constraint is added to the system such that $x - y = 3$. Determine the linear relationship that must exist between the parameters λ and μ to satisfy this constraint.
- Using your result from part (c), determine the unique solution for the coordinates (x, y, z, w) when $w = 2$.

- Hint:**
- **Part (a):** Write the system as an augmented matrix $[A|B]$. Notice that adding the first two rows together creates a row that has the exact same variable coefficients as the third row. For the system to be consistent, the constant terms must also match.
 - **Part (b):** Since the third equation is redundant when consistent, use only the first two equations. Substitute $z = \lambda$ and $w = \mu$, moving them to the right-hand side. Then use simultaneous elimination on x and y .
 - **Part (c):** Substitute your parametric expressions for x and y from part (b) into the equation $x - y = 3$ and group the like parameter terms together.
 - **Part (d):** Substitute $w = \mu = 2$ into your relationship from part (c) to find the value of λ (which represents z). Then, plug both parameter values back into your expressions for x and y .

Solution 4.12

(a)

The augmented matrix is:

$$\left[\begin{array}{cccc|c} 1 & 2 & -1 & 1 & 5 \\ 2 & -1 & 3 & -2 & 0 \\ 3 & 1 & 2 & -1 & k \end{array} \right]$$

Performing the row operation $R_1 + R_2 \rightarrow R_{new}$:

$$(1+2)x + (2-1)y + (-1+3)z + (1-2)w = 5+0 \implies 3x + y + 2z - w = 5.$$

Comparing this to Row 3 ($3x + y + 2z - w = k$), we can see that for the system to be mathematically consistent, $k = 5$.

(b)

Let $z = \lambda$ and $w = \mu$. The system reduces to the first two equations:

$$(1) \quad x + 2y = 5 + \lambda - \mu$$

$$(2) \quad 2x - y = -3\lambda + 2\mu$$

Multiply (2) by 2: $4x - 2y = -6\lambda + 4\mu$.

$$\text{Add to (1): } 5x = 5 - 5\lambda + 3\mu \implies x = 1 - \lambda + \frac{3}{5}\mu.$$

$$\text{Multiply (1) by 2: } 2x + 4y = 10 + 2\lambda - 2\mu.$$

$$\text{Subtract (2) from this: } 5y = 10 + 5\lambda - 4\mu \implies y = 2 + \lambda - \frac{4}{5}\mu.$$

(c)

Apply the constraint: $x - y = 3$.

$$(1 - \lambda + \frac{3}{5}\mu) - (2 + \lambda - \frac{4}{5}\mu) = 3$$

$$-1 - 2\lambda + \frac{7}{5}\mu = 3$$

$$-2\lambda + \frac{7}{5}\mu = 4.$$

Multiplying by 5 yields the linear relationship: $-10\lambda + 7\mu = 20$.

(d)

Given $w = 2$, we have $\mu = 2$.

Substitute into the relationship from (c):

$$-10\lambda + 7(2) = 20 \implies -10\lambda + 14 = 20 \implies -10\lambda = 6 \implies \lambda = -\frac{3}{5}.$$

So, $z = -0.6$.

$$\text{Find } x: x = 1 - (-\frac{3}{5}) + \frac{3}{5}(2) = 1 + \frac{3}{5} + \frac{6}{5} = \frac{14}{5} = 2.8.$$

$$\text{Find } y: y = 2 + (-\frac{3}{5}) - \frac{4}{5}(2) = 2 - \frac{3}{5} - \frac{8}{5} = -\frac{1}{5} = -0.2.$$

The unique solution is $(x, y, z, w) = (\frac{14}{5}, -\frac{1}{5}, -\frac{3}{5}, 2)$ or $(2.8, -0.2, -0.6, 2)$.

Takeaways 4.12

- **Underdetermined Systems:** Matrices representing systems with fewer independent equations than variables yield infinite families of solutions requiring parameters (λ, μ) .
- **Row Operations for Consistency:** Elementary row operations on augmented matrices quickly reveal inconsistencies (e.g., $0 = c$ where $c \neq 0$). If a row reduces entirely to zero on the left, the right side must also be zero for a solution to exist.
- **Applying Constraints:** Adding constraints to a parametric solution family reduces its degrees of freedom. Each new valid equation removes one parameter, eventually leading to a single unique point if enough constraints are provided.

Problem 4.13: Advanced Matrix Products and Inverses

Let $C = \begin{bmatrix} 2 & -1 & 1 \\ 1 & 0 & 2 \\ -1 & 1 & -1 \end{bmatrix}$ and $D = \begin{bmatrix} -2 & 0 & -2 \\ -1 & -1 & -3 \\ 1 & -1 & 1 \end{bmatrix}$.

- (a) Evaluate the matrix product CD . Hence, determine the inverse matrices C^{-1} and D^{-1} in terms of C and D .
- (b) A system of simultaneous linear equations is given by:

$$\begin{aligned} -2x - 2z &= 4k \\ -x - y - 3z &= 2 \\ x - y + z &= -2k \end{aligned}$$

Express this system in the form $DX = V$, and hence solve for x, y , and z in terms of the real constant k .

- (c) Determine the specific value of k such that the solution point (x, y, z) lies on the plane defined by $x + z = 4$.
- (d) Using the inverse properties established in part (a), solve the matrix equation $CYD = I$ for the 3×3 matrix Y , where I is the identity matrix.

Hint:

- **Part (a):** Multiply rows of C by columns of D . The result will be a scalar multiple of the identity matrix, mI . Use the definition of an inverse ($MM^{-1} = I$) to isolate I .
- **Part (b):** The coefficient matrix of the system matches matrix D . To solve $DX = V$, pre-multiply both sides by D^{-1} . Substitute your expression for D^{-1} from part (a) to avoid recalculating an inverse from scratch.
- **Part (c):** Substitute the expressions for x and z (which you found in terms of k in part b) into the plane equation and solve for k .
- **Part (d):** Do not attempt to define Y with 9 unknown variables. Instead, isolate Y algebraically by pre-multiplying by C^{-1} and post-multiplying by D^{-1} . Remember that a matrix commutes with its own inverse (e.g., $CD = mI \Rightarrow DC = mI$).

Solution 4.13**(a)**

$$CD = \begin{bmatrix} 2 & -1 & 1 \\ 1 & 0 & 2 \\ -1 & 1 & -1 \end{bmatrix} \begin{bmatrix} -2 & 0 & -2 \\ -1 & -1 & -3 \\ 1 & -1 & 1 \end{bmatrix} = \begin{bmatrix} -4+1+1 & 0+1-1 & -4+3+1 \\ -2+0+2 & 0+0-2 & -2+0+2 \\ 2-1-1 & 1-1+1 & 2-3-1 \end{bmatrix}$$

$$CD = \begin{bmatrix} -2 & 0 & 0 \\ 0 & -2 & 0 \\ 0 & 0 & -2 \end{bmatrix} = -2I.$$

Since $CD = -2I$, we have $C(-\frac{1}{2}D) = I$, so $C^{-1} = -\frac{1}{2}D$.

Similarly, $(-\frac{1}{2}C)D = I$, so $D^{-1} = -\frac{1}{2}C$.

(b)

The system is $D \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 4k \\ 2 \\ -2k \end{bmatrix}$.

$$X = D^{-1}V = -\frac{1}{2}CV = -\frac{1}{2} \begin{bmatrix} 2 & -1 & 1 \\ 1 & 0 & 2 \\ -1 & 1 & -1 \end{bmatrix} \begin{bmatrix} 4k \\ 2 \\ -2k \end{bmatrix}$$

$$X = -\frac{1}{2} \begin{bmatrix} 2(4k) - 1(2) + 1(-2k) \\ 1(4k) + 0(2) + 2(-2k) \\ -1(4k) + 1(2) - 1(-2k) \end{bmatrix} = -\frac{1}{2} \begin{bmatrix} 6k - 2 \\ 0 \\ -2k + 2 \end{bmatrix} = \begin{bmatrix} 1 - 3k \\ 0 \\ k - 1 \end{bmatrix}.$$

Thus, $x = 1 - 3k$, $y = 0$, and $z = k - 1$.

(c)

We require $x + z = 4$.

Substituting the results from (b):

$$(1 - 3k) + (k - 1) = 4$$

$$-2k = 4 \Rightarrow k = -2.$$

(d)

Given $CYD = I$.

Pre-multiply by C^{-1} and post-multiply by D^{-1} :

$$Y = C^{-1}ID^{-1} = C^{-1}D^{-1}.$$

Substitute the inverses from part (a):

$$Y = (-\frac{1}{2}D)(-\frac{1}{2}C) = \frac{1}{4}DC.$$

Since $CD = -2I$, it follows that $DC = -2I$ (a matrix and its inverse commute).

$$Y = \frac{1}{4}(-2I) = -\frac{1}{2}I = \begin{bmatrix} -0.5 & 0 & 0 \\ 0 & -0.5 & 0 \\ 0 & 0 & -0.5 \end{bmatrix}.$$

Takeaways 4.13

- **The “Hence” Method:** Calculating 3×3 inverses from scratch using determinants and adjugates is tedious. Finding a product $AB = mI$ is a standard shortcut to define $A^{-1} = \frac{1}{m}B$.
- **Parametric Systems:** Matrices handle parametric constants (like k) effortlessly because matrix multiplication distributes linearly over expressions.
- **Matrix Algebra vs. Arithmetic:** Part (d) demonstrates the power of abstract matrix algebra. Treating matrices as single algebraic entities (C^{-1} , D^{-1}) avoids massive blocks of arithmetic calculation.

Problem 4.14: Matrix Systems and Planes in 3D Space

Consider the points $A(1, 2, -1)$, $B(2, 3, 1)$, and $C_k(3, -1, k)$, where k is a real constant. A plane containing these three points can be modeled by the equation $ax + by + cz = 1$, provided the plane does not pass through the origin.

- Substitute the coordinates of the three points into the plane equation to formulate a system of three linear equations in terms of a , b , and c . Express this system as a matrix equation $M\mathbf{v} = \mathbf{u}$, and find $\det(M)$ in terms of k .
- Determine the value of k for which the matrix M is singular. Explain why the matrix equation $M\mathbf{v} = \mathbf{u}$ has no solution for this value of k , and what this implies about the geometric position of the plane containing A , B , and C_k relative to the origin.
- Let $k = 2$, corresponding to the specific point $C_2(3, -1, 2)$. Solve the system using Gaussian elimination (row reduction) to find the values of a , b , and c , and hence state the unique equation of the plane in the form $ax + by + cz + d = 0$ with integer coefficients.
- For the singular case where $k = 18$, the plane must pass through the origin. Set up a new homogeneous matrix equation for the plane $a'x + b'y + c'z = 0$, and find the specific Cartesian equation of this plane by letting $a' = 5$.

Hint:

- **Part (a):** Replace $x, y,$ and z with the coordinates of each point to create three distinct equations. Extract the coefficients of $a, b,$ and c to form the 3×3 matrix M . Calculate the determinant by expanding across the top row.
- **Part (b):** A matrix is singular when its determinant equals zero. If setting $ax + by + cz = 1$ yields a singular matrix and an inconsistent system, it means no such plane exists that misses the origin.
- **Part (c):** Form the augmented matrix $[M|\mathbf{u}]$ and use elementary row operations to reach upper triangular form. Back-substitute to find $c, b,$ and a . Multiply the entire plane equation by the lowest common denominator to remove fractions.
- **Part (d):** A plane through the origin equates to 0 rather than 1. Substitute the points $A, B,$ and C_{18} to form a homogeneous system $M\mathbf{v} = \mathbf{0}$. Use the first two equations to express b' and c' strictly in terms of a' , then substitute $a' = 5$.

Solution 4.14**(a)**Substituting the points into $ax + by + cz = 1$:

$$A(1, 2, -1) \implies a + 2b - c = 1$$

$$B(2, 3, 1) \implies 2a + 3b + c = 1$$

$$C_k(3, -1, k) \implies 3a - b + kc = 1$$

$$\text{Matrix equation: } \begin{bmatrix} 1 & 2 & -1 \\ 2 & 3 & 1 \\ 3 & -1 & k \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}.$$

$$\det(M) = 1(3k - (-1)) - 2(2k - 3) + (-1)(-2 - 9) = 3k + 1 - 4k + 6 + 11 = 18 - k.$$

(b)Matrix M is singular when $\det(M) = 0 \implies 18 - k = 0 \implies k = 18$.

When $k = 18$, the system is inconsistent (parallel planes in the row space), meaning no unique solution exists for $ax + by + cz = 1$. Geometrically, this implies the plane containing A, B , and C_{18} must pass exactly through the origin, meaning its true equation is of the form $ax + by + cz = 0$, not 1.

(c)Let $k = 2$. The augmented matrix is:

$$\left[\begin{array}{ccc|c} 1 & 2 & -1 & 1 \\ 2 & 3 & 1 & 1 \\ 3 & -1 & 2 & 1 \end{array} \right]$$

$$R_2 \rightarrow R_2 - 2R_1 \implies \left[\begin{array}{ccc|c} 1 & 2 & -1 & 1 \\ 0 & -1 & 3 & -1 \\ 3 & -1 & 2 & 1 \end{array} \right]$$

$$R_3 \rightarrow R_3 - 3R_1 \implies \left[\begin{array}{ccc|c} 1 & 2 & -1 & 1 \\ 0 & -1 & 3 & -1 \\ 0 & -7 & 5 & -2 \end{array} \right]$$

$$R_3 \rightarrow R_3 - 7R_2 \implies \left[\begin{array}{ccc|c} 1 & 2 & -1 & 1 \\ 0 & -1 & 3 & -1 \\ 0 & 0 & -16 & 5 \end{array} \right]$$

$$\text{From } R_3: -16c = 5 \implies c = -\frac{5}{16}.$$

$$\text{From } R_2: -b + 3(-\frac{5}{16}) = -1 \implies b = \frac{1}{16}.$$

$$\text{From } R_1: a + 2(\frac{1}{16}) - (-\frac{5}{16}) = 1 \implies a = \frac{9}{16}.$$

The equation is $\frac{9}{16}x + \frac{1}{16}y - \frac{5}{16}z = 1$, which simplifies to $9x + y - 5z - 16 = 0$.

(d)For $k = 18$, the homogeneous system is:

$$(1) a' + 2b' - c' = 0$$

$$(2) 2a' + 3b' + c' = 0$$

$$(3) 3a' - b' + 18c' = 0$$

$$\text{Add (1) and (2) to eliminate } c': 3a' + 5b' = 0 \implies b' = -\frac{3}{5}a'.$$

$$\text{Let } a' = 5 \implies b' = -3.$$

$$\text{Substitute into (1): } 5 + 2(-3) - c' = 0 \implies -1 - c' = 0 \implies c' = -1.$$

The equation of the plane is $5x - 3y - z = 0$.

Takeaways 4.14

- **Non-Homogeneous vs Homogeneous Form:** Setting a plane's constant d to a non-zero value allows for a quick 3×3 matrix setup, but it operates on the assumption that the plane does not cross the origin.
- **Geometric Meaning of Singularity:** If the resulting matrix determinant is zero, it algebraically signals that the initial assumption (the plane avoids the origin) is false for that specific set of points.
- **Gaussian Elimination:** Row reduction is a powerful tool for solving 3×3 systems without needing to calculate cumbersome matrix inverses or adjugates manually.

Problem 4.15: Matrix Polynomials and System Solutions

Let $A = \begin{bmatrix} 3 & -2 & -2 \\ 3 & -2 & -3 \\ 1 & -1 & 0 \end{bmatrix}$.

- Evaluate A^2 to show that A is an involutory matrix (i.e., a matrix where $A^2 = I$).
- Let $E = \frac{1}{2}(I + A)$. Using matrix algebra and your result from part (a), prove that E is an idempotent matrix (i.e., $E^2 = E$).
- Consider the matrix $M = I + kA$, where k is a real constant. By expanding the algebraic product $M(I - kA)$, determine the exact values of k for which the matrix M does not have an inverse.
- Let $k = 2$. Use your algebraic result from part (c) to determine M^{-1} , and hence solve the following system of linear equations:

$$\begin{aligned} 7x - 4y - 4z &= 3 \\ 6x - 3y - 6z &= -3 \\ 2x - 2y + z &= 6 \end{aligned}$$

Hint:

- **Part (a):** Multiply A by itself using standard row-by-column multiplication. Verify that the main diagonal consists of 1s and all other entries are 0.
- **Part (b):** Square the algebraic expression for E . Remember that scalar multiples can be factored out, and matrix multiplication distributes over addition. Substitute $A^2 = I$ when simplifying.
- **Part (c):** Expand $(I + kA)(I - kA)$ as you would a difference of perfect squares, keeping in mind that I commutes with any matrix. For M to be singular, the scalar coefficient of I in your resulting expression must equal zero.
- **Part (d):** Notice that the coefficient matrix of the system is precisely M when $k = 2$. Calculate M^{-1} by setting $k = 2$ in the identity from part (c), then pre-multiply the constant column vector by M^{-1} .

Solution 4.15

(a)

$$A^2 = \begin{bmatrix} 3 & -2 & -2 \\ 3 & -2 & -3 \\ 1 & -1 & 0 \end{bmatrix} \begin{bmatrix} 3 & -2 & -2 \\ 3 & -2 & -3 \\ 1 & -1 & 0 \end{bmatrix}$$

$$A^2 = \begin{bmatrix} 9-6-2 & -6+4+2 & -6+6+0 \\ 9-6-3 & -6+4+3 & -6+6+0 \\ 3-3+0 & -2+2+0 & -2+3+0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} = I.$$

(b)

$$E^2 = \left(\frac{1}{2}(I+A)\right)^2 = \frac{1}{4}(I+A)(I+A)$$

$$E^2 = \frac{1}{4}(I^2 + IA + AI + A^2).$$

Since $I^2 = I$, $IA = AI = A$, and $A^2 = I$:

$$E^2 = \frac{1}{4}(I + A + A + I) = \frac{1}{4}(2I + 2A) = \frac{1}{2}(I + A) = E.$$

(c)

$$M(I - kA) = (I + kA)(I - kA) = I^2 - kIA + kAI - k^2A^2.$$

Because $IA = AI = A$, the middle terms cancel:

$$M(I - kA) = I - k^2A^2.$$

Substitute $A^2 = I$:

$$M(I - kA) = I - k^2I = (1 - k^2)I.$$

$$\text{Rearranging this gives } M \left[\frac{1}{1-k^2}(I - kA) \right] = I.$$

Matrix M is singular (has no inverse) when the scalar divisor $1 - k^2 = 0$.

Therefore, M cannot have an inverse when $k = 1$ or $k = -1$.

(d)

Let $k = 2$. The system's coefficient matrix is $M = I + 2A = \begin{bmatrix} 7 & -4 & -4 \\ 6 & -3 & -6 \\ 2 & -2 & 1 \end{bmatrix}$.

From (c), $M^{-1} = \frac{1}{1-2^2}(I - 2A) = -\frac{1}{3}(I - 2A)$.

$$I - 2A = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} - \begin{bmatrix} 6 & -4 & -4 \\ 6 & -4 & -6 \\ 2 & -2 & 0 \end{bmatrix} = \begin{bmatrix} -5 & 4 & 4 \\ -6 & 5 & 6 \\ -2 & 2 & 1 \end{bmatrix}.$$

$$M^{-1} = -\frac{1}{3} \begin{bmatrix} -5 & 4 & 4 \\ -6 & 5 & 6 \\ -2 & 2 & 1 \end{bmatrix}.$$

$$X = M^{-1}B = -\frac{1}{3} \begin{bmatrix} -5 & 4 & 4 \\ -6 & 5 & 6 \\ -2 & 2 & 1 \end{bmatrix} \begin{bmatrix} 3 \\ -3 \\ 6 \end{bmatrix}$$

$$X = -\frac{1}{3} \begin{bmatrix} -15 - 12 + 24 \\ -18 - 15 + 36 \\ -6 - 6 + 6 \end{bmatrix} = -\frac{1}{3} \begin{bmatrix} -3 \\ 3 \\ -6 \end{bmatrix} = \begin{bmatrix} 1 \\ -1 \\ 2 \end{bmatrix}.$$

Therefore, $x = 1$, $y = -1$, and $z = 2$.

Takeaways 4.15

- **Abstract Matrix Algebra:** Abstract algebra simplifies complex computational problems. Establishing $A^2 = I$ creates a generalized shortcut for inverting any matrix of the parametric form $I + kA$.
- **Polynomial Singularity Proofs:** The QCAA curriculum emphasizes proving invertibility using polynomial identities (like $(I + kA)(I - kA) = (1 - k^2)I$) rather than relying exclusively on calculating massive 3×3 determinants.
- **Idempotent and Involutory Links:** Idempotent matrices ($E^2 = E$, which geometrically represent projections) and involutory matrices ($A^2 = I$, which geometrically represent reflections) are fundamentally linked via simple algebraic transformations like $E = \frac{1}{2}(I + A)$.

5 Conclusion

Matrices provide a powerful language for describing and solving complex systems. As this booklet grows, it will become a compact reference for matrix operations and transformations.

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A Appendices

A.1 Further Reading in Linear Algebra